Lecture Notes

Wavelet Theory

From L^2 Foundations to Nonlinear Approximation Theory, Filter Banks, and Operator Theory

Functional Analysis · MRA · Daubechies · Frames · CWT · Besov Spaces

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ABSTRACT

This set of lecture notes develops wavelet theory from first principles, with every proof written out in full and accompanied by figures. We begin with the $L^2(\mathbb{R})$ Hilbert space theory, proving Riesz–Fischer, the projection theorem, Parseval–Bessel, and the Duffin–Schaeffer frame theorem, including the spectral properties of the frame operator. The Fourier transform is extended via the Schwartz class and the Plancherel isometry; the Poisson summation formula is proved in full and deployed throughout. The Haar system's completeness is established via the Lebesgue differentiation theorem. Multiresolution analysis (MRA) is developed from its axioms through the two-scale relation, the QMF condition (derived via Poisson summation), orthogonal complement spaces, and the existence and uniqueness of the mother wavelet. The Daubechies D_{2N} family is constructed by solving the Bézout equation in the ring of trigonometric polynomials, with a full analysis of regularity via the infinite product representation of $\widehat{\varphi}$; a classical symmetry obstruction shows why no compactly supported orthonormal wavelet other than Haar can be symmetric, motivating the biorthogonal (Cohen–Daubechies–Feauveau) alternative. The continuous wavelet transform is characterised by the Calderón–Grossmann–Morlet theorem; a complete functional-analytic proof of the inversion formula is given. The discrete wavelet transform is analysed via polyphase z-transforms, perfect reconstruction conditions, and Sweldens's lifting factorisation. We then develop Besov and Triebel–Lizorkin spaces via wavelet characterisations, nonlinear approximation theory in these spaces, and the Donoho–Johnstone theory of wavelet shrinkage. Applications to JPEG 2000, compressed sensing, and turbulence analysis are discussed. All constructions are illustrated with figures.

CONTENTS

1. Hilbert Space Theory: Complete Foundations

Wavelet theory is, at bottom, a chapter of Hilbert space theory: every construction to follow — the Haar system, multiresolution analysis, the continuous and discrete wavelet transforms — is an orthonormal or frame expansion in $L^2(\mathbb{R})$, and every proof of convergence, completeness, or stability reduces eventually to the geometry of that single Hilbert space. It is therefore worth pausing, before any mention of a wavelet, to develop this geometry with complete rigour: the projection theorem, which underlies the very notion of "best approximation" at a given scale; Parseval's theorem, which converts orthonormal expansions into an exact bookkeeping of energy; and the theory of frames, which will later let us relax orthonormality without losing stable reconstruction. Readers already fluent in Hilbert space theory may treat this section as a reference and skip ahead to Section 2; everyone else should read it in full, since the arguments given here — the three-step subsequence construction in Riesz–Fischer, the parallelogram trick in the projection theorem — recur, essentially unchanged, throughout the rest of these notes.

Observation 1.1 (Historical note)**.** Riesz and Fischer proved completeness of L^2 independently in 1907, in the immediate aftermath of Hilbert's own 1904–1910 work on integral equations and quadratic forms in infinitely many variables — the setting in which the term "Hilbert space" would later be coined by others in his honour. Frame theory is younger: Duffin and Schaeffer introduced frames in 1952 to study nonharmonic Fourier series, decades before Daubechies, Grossmann, and Meyer revived the concept in the mid-1980s for exactly the wavelet applications developed below.

1.1. Measure Theory Prerequisites

We work on $(\mathbb{R}, \mathcal{B}, \lambda)$ where $\mathcal{B}$ is the Borel σ-algebra and λ is Lebesgue measure. The following facts are used without proof but stated for precision.

Theorem 1.2 (Lebesgue Dominated Convergence)**.** *If $f_n \to f$ a.e. and $|f_n| \leq g \in L^1(\mathbb{R})$ for all n, then $f \in L^1(\mathbb{R})$ and $\|f_n - f\|_{L^1} \to 0$.*

Theorem 1.3 (Lebesgue Differentiation)**.** *For $f \in L^1_{\mathrm{loc}}(\mathbb{R})$, for a.e. $x \in \mathbb{R}$:*

$$f(x) = \lim_{r \to 0^+} \frac{1}{2r} \int_{x-r}^{x+r} f(t)\, \mathrm{d}t.$$

Theorem 1.4 (Fubini–Tonelli)**.** *If $F : \mathbb{R}^2 \to \mathbb{C}$ is measurable and $\int_{\mathbb{R}} \int_{\mathbb{R}} |F(x,y)|\, \mathrm{d}x\, \mathrm{d}y < \infty$, then the iterated integrals equal the double integral, in any order.*

1.2. $L^2(\mathbb{R})$ as a Hilbert Space

Definition 1.5. $L^2(\mathbb{R}) := \{f : \mathbb{R} \to \mathbb{C} \text{ measurable} : \|f\|^2 := \int_{\mathbb{R}} |f(x)|^2 \,\mathrm{d}x < \infty\}/\sim$, where $f \sim g \iff f = g$ a.e. The inner product is $\langle f, g\rangle := \int_{\mathbb{R}} f(x)\overline{g(x)}\mathrm{d}x$.

Lemma 1.6 (Cauchy–Schwarz)**.** $|\langle f, g\rangle| \leq \|f\|\,\|g\|$, *with equality iff* $f = cg$ *a.e. for some* $c \in \mathbb{C}$.

Proof. For $g \neq 0$, write $f = (f - \frac{\langle f, g\rangle}{\|g\|^2}g) + \frac{\langle f, g\rangle}{\|g\|^2}g$. The two summands are orthogonal, so by Pythagoras: $\|f\|^2 = \left\|f - \frac{\langle f, g\rangle}{\|g\|^2}g\right\|^2 + \frac{|\langle f, g\rangle|^2}{\|g\|^2} \geq \frac{|\langle f, g\rangle|^2}{\|g\|^2}$. Equality holds iff the first term vanishes, i.e. $f = \frac{\langle f, g\rangle}{\|g\|^2}g$ a.e. □

Lemma 1.7 (Minkowski Integral Inequality)**.** *For measurable* $F(x, t) \geq 0$: $\left\|\int F(\cdot, t)\mathrm{d}t\right\|_{L^2} \leq \int \|F(\cdot, t)\|_{L^2}\,\mathrm{d}t$.

Proof. Let $G(x) = \int F(x, t)\mathrm{d}t$. By Tonelli (to justify the exchange), then Hölder with exponents 2 and 2:

$$\begin{aligned}\|G\|_{L^2}^2 &= \int_{\mathbb{R}} G(x)^2\mathrm{d}x = \int_{\mathbb{R}} G(x)\int_{\mathbb{R}} F(x, t)\mathrm{d}t\,\mathrm{d}x \\ &\overset{\text{Fubini}}{=} \int_{\mathbb{R}}\int_{\mathbb{R}} G(x)F(x, t)\mathrm{d}x\,\mathrm{d}t \leq \int_{\mathbb{R}} \|G\|_{L^2}\,\|F(\cdot, t)\|_{L^2}\,\mathrm{d}t = \|G\|_{L^2}\int_{\mathbb{R}} \|F(\cdot, t)\|_{L^2}\,\mathrm{d}t.\end{aligned}$$

Divide by $\|G\|_{L^2} > 0$. □

Theorem 1.8 (Riesz–Fischer: Complete Proof)**.** $(L^2(\mathbb{R}), \langle\cdot, \cdot\rangle)$ *is a complete, separable Hilbert space.*

Proof. **(I) Inner product.** Sesquilinearity and conjugate-symmetry are immediate. Positive-definiteness: $\|f\|^2 = 0 \Rightarrow \int |f|^2 = 0 \Rightarrow |f|^2 = 0$ a.e. $\Rightarrow f = 0$ in $L^2(\mathbb{R})$. Well-definedness: if $f = g$ a.e. then $\langle f, \cdot\rangle = \langle g, \cdot\rangle$.

(II) Completeness. Let $(f_n)_{n\geq 1}$ be Cauchy. *Step 1*: extract a rapidly Cauchy subsequence. For each $k \geq 1$, the Cauchy property yields N_k such that $\|f_m - f_n\| < 2^{-k}$ for $m, n \geq N_k$; set $n_k = \max(N_k, n_{k-1} + 1)$, giving $\left\|f_{n_{k+1}} - f_{n_k}\right\| < 2^{-k}$.

Step 2: Construct the limit. Define

$$G_M(x) := \sum_{k=1}^{M} \left|f_{n_{k+1}}(x) - f_{n_k}(x)\right|, \quad G(x) := \sum_{k=1}^{\infty} \left|f_{n_{k+1}}(x) - f_{n_k}(x)\right|.$$

By Minkowski's inequality (Lemma 1.7) applied to the finite sum, then monotone convergence as $M \to \infty$:

$$\|G\|_{L^2} = \lim_{M\to\infty} \|G_M\|_{L^2} \le \sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\| < \sum_{k=1}^{\infty} 2^{-k} = 1.$$

So $G \in L^2(\mathbb{R})$, hence $G(x) < \infty$ a.e. At every such x, the series $\sum_k (f_{n_{k+1}}(x) - f_{n_k}(x))$ converges absolutely, so

$$f(x) := f_{n_1}(x) + \sum_{k=1}^{\infty} \big(f_{n_{k+1}}(x) - f_{n_k}(x)\big) = \lim_{K\to\infty} f_{n_K}(x)$$

is well-defined a.e., and $|f_{n_K}(x)| \le |f_{n_1}(x)| + G(x) \in L^2$ for all K.

Step 3: Show $f_{n_k} \to f$ in L^2. Since $f_{n_K}(x) \to f(x)$ a.e. and $|f_{n_K} - f|^2 \le (|f_{n_K}| + |f|)^2 \le 4(|f_{n_1}| + G)^2 \in L^1$, the dominated convergence theorem (1.2) gives $\|f_{n_K} - f\| \to 0$.

Step 4: The full sequence converges. Given $\varepsilon > 0$, find N with $\|f_m - f_n\| < \varepsilon/2$ for $m, n \ge N$, and K with $n_K \ge N$ and $\|f_{n_K} - f\| < \varepsilon/2$. Then for $n \ge N$: $\|f_n - f\| \le \|f_n - f_{n_K}\| + \|f_{n_K} - f\| < \varepsilon$.

(III) Separability. Let $\mathcal{D}$ be the set of finite linear combinations $\sum_{j=1}^{m} (a_j + ib_j)\mathbf{1}_{[p_j,q_j)}$ with $a_j, b_j, p_j, q_j \in \mathbb{Q}$. This is countable. Given $f \in L^2(\mathbb{R})$ and $\varepsilon > 0$:

(a) Find $f_R = f \cdot \mathbf{1}_{[-R,R]}$ with $\|f - f_R\| < \varepsilon/3$ (DCT).
(b) Find $f_{R,M} = \min(|f_R|, M) \cdot \mathrm{sgn}(f_R)$ bounded by M; $\|f_R - f_{R,M}\| < \varepsilon/3$.
(c) Approximate $f_{R,M}$ (bounded, compactly supported) by a step function $s \in \mathcal{D}$: choose a fine enough partition of $[-R, R]$ and use density of $\mathbb{Q}$ in $\mathbb{R}$, so $\|f_{R,M} - s\| < \varepsilon/3$.

Then $\|f - s\| < \varepsilon$. □

Observation 1.9 (Why completeness is non-trivial). $L^2(\mathbb{R})$ is fundamentally different from $C_c(\mathbb{R})$ (continuous compactly supported functions), which is *not* complete: the step function limit of smooth approximations to $\mathbf{1}_{[0,1]}$ is not continuous. Completeness is the reason we can take limits of sequences of approximating subspaces in MRA and be guaranteed the limits lie in $L^2(\mathbb{R})$. It is also why the projection theorem holds for *closed* subspaces—without completeness, the infimum of a minimisation problem might not be attained.

1.3. Orthogonal Projections and the Projection Theorem

Theorem 1.10 (Projection Theorem). *Let $V \subset L^2(\mathbb{R})$ be a closed subspace. Then:*

(i) $L^2(\mathbb{R}) = V \oplus V^\perp$ *(orthogonal direct sum), where* $V^\perp := \{f \in L^2(\mathbb{R}) : \langle f, v\rangle = 0\ \forall v \in V\}$.

(ii) *There exists a unique bounded linear operator* $P_V : L^2(\mathbb{R}) \to V$ *(the* orthogonal projection*) with* $P_V^2 = P_V$*,* $P_V^* = P_V$*,* $\|P_V\| = 1$*, and* $\ker P_V = V^\perp$.

(iii) $P_V f$ *is the unique minimiser of* $\|f - v\|$ *over* $v \in V$.

Proof. **Existence and uniqueness of $P_V f$.** Let $d := \inf_{v\in V} \|f - v\| \geq 0$ and choose $(v_n) \subset V$ with $\|f - v_n\| \to d$. Apply the parallelogram law to $f - v_n$ and $f - v_m$:

$$\|v_n - v_m\|^2 = 2\|f - v_n\|^2 + 2\|f - v_m\|^2 - 4\left\|f - \frac{v_n + v_m}{2}\right\|^2.$$

Since V is a subspace, $\frac{v_n+v_m}{2} \in V$, so $\|f - \frac{v_n+v_m}{2}\| \geq d$, giving $\|v_n - v_m\|^2 \leq 2\|f - v_n\|^2 + 2\|f - v_m\|^2 - 4d^2 \to 0$. So (v_n) is Cauchy; since V is closed, $v_n \to P_V f \in V$. Uniqueness: if $\|f - w\| = d$ for another $w \in V$, then by the same parallelogram argument $\|P_V f - w\| = 0$.

Orthogonality. For any $v \in V$ and $t \in \mathbb{R}$: $\|f - P_V f - tv\|^2 \geq d^2 = \|f - P_V f\|^2$, expanding: $-2t\,\mathrm{Re}\langle f - P_V f, v\rangle + t^2\|v\|^2 \geq 0$ for all $t \in \mathbb{R}$. The coefficient of t must vanish (if not, take t small and of the right sign to make the expression negative). Hence $\mathrm{Re}\langle f - P_V f, v\rangle = 0$. Replacing $v \to iv$ gives $\mathrm{Im}\langle f - P_V f, v\rangle = 0$, so $\langle f - P_V f, v\rangle = 0$. Thus $f - P_V f \in V^\perp$.

Direct sum. Any $f = P_V f + (f - P_V f) \in V + V^\perp$. If $f \in V \cap V^\perp$: $\langle f, f\rangle = 0$, so $f = 0$.

Linearity and boundedness of P_V. $P_V(f+g) = P_V f + P_V g$ follows from uniqueness of the decomposition. $\|P_V f\|^2 + \|f - P_V f\|^2 = \|f\|^2$ (Pythagoras), so $\|P_V f\| \leq \|f\|$ ($\|P_V\| \leq 1$; equality at $f \in V$). Idempotency: $P_V(P_V f) = P_V f$ since $P_V f \in V$. Self-adjointness: $\langle P_V f, g\rangle = \langle P_V f, P_V g\rangle = \langle f, P_V g\rangle$. ☐

Observation 1.11 (MRA projections as approximations)**.** In MRA, $P_j := P_{V_j}$ denotes the projection onto the j-th level. The sequence $P_j f \to f$ as $j \to +\infty$ (density, MRA2) and $P_j f \to 0$ as $j \to -\infty$ (separation, MRA3) encodes the idea that finer scales capture more detail. The *detail* captured at level j is $Q_j f := (P_{j+1} - P_j) f = P_{W_j} f$, which is the projection onto the wavelet subspace $W_j = V_{j+1} \ominus V_j$. The total energy satisfies $\|f\|^2 = \sum_{j\in\mathbb{Z}} \|Q_j f\|^2$, a wavelets-based energy decomposition.

1.4. Orthonormal Bases and Parseval's Theorem

Theorem 1.12 (Parseval–Bessel)**.** *Let* $\{e_n\}_{n\in\mathbb{Z}} \subset L^2(\mathbb{R})$ *be an ONB. Then for all* $f, g \in L^2(\mathbb{R})$*:*

(i) $\sum_n |\langle f, e_n\rangle|^2 \le \|f\|^2$ *(Bessel, for any ON set).*

(ii) $f = \sum_n \langle f, e_n\rangle e_n$ *(norm convergence)* *(completeness $\Leftrightarrow$ equality in (i)).*

(iii) $\langle f, g\rangle = \sum_n \langle f, e_n\rangle \overline{\langle g, e_n\rangle}$ *(Parseval).*

(iv) $\|f\|^2 = \sum_n |\langle f, e_n\rangle|^2$ *(Plancherel for ONB).*

Proof. **(i) Bessel's inequality.** Let $S_N = \sum_{|n|\le N} \langle f, e_n\rangle e_n$. Since $f - S_N \perp S_N$ (by orthonormality: $\langle f - S_N, e_m\rangle = 0$ for $|m| \le N$), Pythagoras gives $\|f\|^2 = \|S_N\|^2 + \|f - S_N\|^2 \ge \|S_N\|^2 = \sum_{|n|\le N} |\langle f, e_n\rangle|^2$. Let $N \to \infty$.

(ii) Convergence. The partial sums (S_N) are Cauchy:

$$\|S_M - S_N\|^2 = \sum_{N<|n|\le M} |\langle f, e_n\rangle|^2 \to 0$$

by (i) (the series converges). So $S_N \to \tilde{f} \in L^2(\mathbb{R})$. For any m, by continuity of the inner product:

$$\langle \tilde{f}, e_m\rangle = \lim_N \langle S_N, e_m\rangle = \langle f, e_m\rangle.$$

Hence $\langle f - \tilde{f}, e_m\rangle = 0$ for all m. By completeness of the ONB (spanning condition), $f - \tilde{f} = 0$ a.e., so $f = \tilde{f}$.

(iii)–(iv) follow by bilinearity of the inner product and polarisation: $\langle f, g\rangle = \frac{1}{4}[\|f+g\|^2 - \|f-g\|^2 + i\,\|f+ig\|^2 - i\,\|f-ig\|^2]$. Apply (iv) to each term. $\square$

1.5. Frames: Theory and Spectral Analysis

Definition 1.13. A sequence $\{\psi_\lambda\}_{\lambda\in\Lambda} \subset L^2(\mathbb{R})$ is a *frame* with bounds $0 < A \le B < \infty$ if $A\,\|f\|^2 \le \sum_\lambda |\langle f, \psi_\lambda\rangle|^2 \le B\,\|f\|^2$ for all $f \in L^2(\mathbb{R})$. The *analysis operator* is $T : f \mapsto (\langle f, \psi_\lambda\rangle)_\lambda : L^2(\mathbb{R}) \to \ell^2(\Lambda)$, the *synthesis operator* is $T^* : (c_\lambda) \mapsto \sum_\lambda c_\lambda \psi_\lambda : \ell^2 \to L^2(\mathbb{R})$, and the *frame operator* is $S := T^*T : L^2(\mathbb{R}) \to L^2(\mathbb{R})$, $Sf = \sum_\lambda \langle f, \psi_\lambda\rangle \psi_\lambda$.

Observation 1.14 (Adjoint computation)**.** $(Tf)_\lambda = \langle f, \psi_\lambda\rangle$, and

$$\langle Tf, c\rangle_{\ell^2} = \sum_\lambda \langle f, \psi_\lambda\rangle \overline{c_\lambda} = \langle f, \sum_\lambda c_\lambda \psi_\lambda\rangle_{L^2(\mathbb{R})} = \langle f, T^*c\rangle,$$

confirming that T^* is the synthesis operator. The frame condition then reads $AI \le T^*T \le BI$ in the operator order.

Theorem 1.15 (Duffin–Schaeffer: Full Spectral Analysis)**.** *Let $\{\psi_\lambda\}$ be a frame with bounds A, B.*

(a) *S is self-adjoint, positive, and bounded with $\|S\| \le B$, $\|S^{-1}\| \le A^{-1}$.*

(b) *The spectrum* $\sigma(S) \subset [A, B]$*; in particular* S *is boundedly invertible.*

(c) *The dual frame* $\{S^{-1}\psi_\lambda\}$ *has bounds* B^{-1}, A^{-1} *and satisfies* $f = \sum_\lambda \langle f, \psi_\lambda\rangle S^{-1}\psi_\lambda = \sum_\lambda \langle f, S^{-1}\psi_\lambda\rangle \psi_\lambda$ *for all* $f \in L^2(\mathbb{R})$*.*

(d) *The* canonical tight frame $\{S^{-1/2}\psi_\lambda\}$ *has bound* 1 *(a Parseval frame).*

Proof. **(a)** Self-adjointness: $\langle Sf, g\rangle = \sum_\lambda \langle f, \psi_\lambda\rangle\langle\psi_\lambda, g\rangle = \langle f, Sg\rangle$. The frame inequality reads $A\langle f, f\rangle \le \langle Sf, f\rangle \le B\langle f, f\rangle$, giving $AI \le S \le BI$ as operators ($\langle (S - AI)f, f\rangle \ge 0$). So $\|S - AI\| \le B - A$ and $\|S\| \le B$.

(b) By the spectral theorem for bounded self-adjoint operators on Hilbert space, $\sigma(S) \subset [m, M]$ where $m = \inf_{\|f\|=1}\langle Sf, f\rangle$ and $M = \sup_{\|f\|=1}\langle Sf, f\rangle$. From the frame inequality, $m \ge A > 0$ and $M \le B$. Since $m > 0$, $0 \notin \sigma(S)$, so S is invertible with $\sigma(S^{-1}) = \sigma(S)^{-1} \subset [B^{-1}, A^{-1}]$.

(c) Reconstruction formula: $f = SS^{-1}f = \sum_\lambda \langle S^{-1}f, \psi_\lambda\rangle\psi_\lambda$. Since S is self-adjoint, so is S^{-1}, hence $\langle S^{-1}f, \psi_\lambda\rangle = \langle f, S^{-1}\psi_\lambda\rangle$, giving both reconstruction formulas simultaneously: $f = \sum_\lambda \langle f, \psi_\lambda\rangle S^{-1}\psi_\lambda = \sum_\lambda \langle f, S^{-1}\psi_\lambda\rangle\psi_\lambda$.

Dual frame bounds. The dual analysis operator $T_{\text{dual}} : f \mapsto (\langle f, S^{-1}\psi_\lambda\rangle)_\lambda$ satisfies $T_{\text{dual}} = TS^{-1}$, since $\langle f, S^{-1}\psi_\lambda\rangle = \langle S^{-1}f, \psi_\lambda\rangle$. Hence the dual frame operator is

$$S_{\text{dual}} := T^*_{\text{dual}}T_{\text{dual}} = S^{-1}T^*TS^{-1} = S^{-1}SS^{-1} = S^{-1}.$$

Consequently $\sum_\lambda \left|\langle f, S^{-1}\psi_\lambda\rangle\right|^2 = \langle S^{-1}f, f\rangle$, and since $\sigma(S^{-1}) \subset [B^{-1}, A^{-1}]$ (part (b)), the same spectral-theorem argument used for S in part (a) applies verbatim to S^{-1} and gives directly

$$B^{-1}\|f\|^2 \le \langle S^{-1}f, f\rangle = \sum_\lambda \left|\langle f, S^{-1}\psi_\lambda\rangle\right|^2 \le A^{-1}\|f\|^2.$$

(d) For the canonical tight frame: $\sum_\lambda \left|\langle f, S^{-1/2}\psi_\lambda\rangle\right|^2 = \langle T^*Tf, S^{-1}f\rangle = \langle Sf, S^{-1}f\rangle = \langle f, f\rangle = \|f\|^2$. □

2. THE FOURIER TRANSFORM: RIGOROUS L^2 THEORY

Having built the abstract Hilbert space machinery, we now need one concrete unitary operator on $L^2(\mathbb{R})$ before any wavelet can be written down: the Fourier transform. Its role in what follows is not merely computational. The two-scale relation, the QMF condition, the mother-wavelet construction, the Daubechies regularity estimates, and the Calderón inversion formula are all, in the end, statements about $\widehat{\varphi}$ and

$\widehat{\psi}$; working in frequency turns questions about orthogonality of infinitely many translates and dilates — genuinely hard to see directly in x-space — into pointwise or Fourier-coefficient identities that can be verified by direct computation. The technical heart of this section is the Poisson summation formula, proved here in full; it is the single bridge, used again and again below, that converts a sum over integer translates in x-space into a periodisation in ξ-space, and back.

Observation 2.1 (Historical note)**.** Poisson stated his summation formula in 1827, decades before Cantor's set theory or Lebesgue's measure theory existed to make sense of the convergence involved; Plancherel's theorem dates to 1910, in the same short window as Riesz–Fischer, once $L^2(\mathbb{R})$ itself had been properly defined. That a 19th-century identity about periodic sums would become, 150 years later, the key computational device for verifying orthonormality of wavelet translates is a recurring pattern in this subject: much of the analytic machinery predates wavelets by a century or more, and is simply redeployed.

2.1. The Schwartz Class and Classical Fourier Transform

Definition 2.2 (Schwartz Class)**.** $\mathcal{S}(\mathbb{R}) := \{f \in C^\infty(\mathbb{R}) : \sup_x \left|x^\alpha \partial^\beta f(x)\right| < \infty\ \forall \alpha, \beta \in \mathbb{N}_0\}$.

On $\mathcal{S}(\mathbb{R})$, define $\widehat{f}(\xi) := \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} \mathrm{d}x$. Key properties on $\mathcal{S}$:

- $\mathcal{F} : \mathcal{S} \to \mathcal{S}$ is a bijection (with inverse $\mathcal{F}^{-1} g(x) = \widehat{g}(-x)$).
- $\mathcal{F}[f'](\xi) = 2\pi i \xi \widehat{f}(\xi)$, $\mathcal{F}[xf](\xi) = \frac{-1}{2\pi i} \widehat{f}'(\xi)$.
- $\mathcal{F}[f * g] = \widehat{f}\widehat{g}$ (convolution theorem, proved by Fubini on $\mathcal{S}$).
- $\mathcal{F}^2[f](x) = f(-x)$, hence $\mathcal{F}^4 = \mathrm{id}$.

Theorem 2.3 (Plancherel: Complete Proof)**.** *$\mathcal{F}$ extends uniquely to a unitary operator $\mathcal{F} : L^2(\mathbb{R}) \to L^2(\mathbb{R})$ with $\left\|\widehat{f}\right\| = \|f\|$ and $\langle \widehat{f}, \widehat{g} \rangle = \langle f, g \rangle$ for all $f, g \in L^2(\mathbb{R})$.*

Proof. **Step 1: Isometry on $\mathcal{S}$.** For $f \in \mathcal{S}$, define $\check{f}(x) := \widehat{f}(-x) = \int f(\xi) e^{2\pi i \xi x} \mathrm{d}\xi$ (inverse Fourier transform). Let $h = f * \check{\bar{f}}$ where $\bar{f}(x) = \overline{f(x)}$. Then $\widehat{h} = \widehat{f} \cdot \overline{\widehat{f}} = \left|\widehat{f}\right|^2 \geq 0$, and by the inversion formula (valid on $\mathcal{S} \subset L^1$): $h(0) = \int \widehat{h}(\xi) \mathrm{d}\xi$. But $h(0) = \int f(x)\overline{f(x)} \mathrm{d}x = \|f\|^2$, and $\int \widehat{h} = \int \left|\widehat{f}\right|^2 = \left\|\widehat{f}\right\|^2$. Hence $\left\|\widehat{f}\right\| = \|f\|$ for all $f \in \mathcal{S}$.

Step 2: Extension by the BLT theorem. $\mathcal{S}$ is dense in $L^2(\mathbb{R})$ (since $C_c^\infty \subset \mathcal{S}$ is dense). The Fourier transform on $\mathcal{S}$ is an isometry into $L^2(\mathbb{R})$, so it extends uniquely to a linear isometry $\mathcal{F} : L^2(\mathbb{R}) \to L^2(\mathbb{R})$ (bounded linear transformation theorem: any bounded linear map from a dense subspace of a Banach space extends uniquely to the whole space, preserving the norm bound).

Step 3: Surjectivity. $\mathcal{F}(\mathcal{S}) = \mathcal{S}$ (since $\mathcal{F}^4 = \mathrm{id}$ on $\mathcal{S}$, $\mathcal{F}$ is bijective there). Hence $\mathcal{F}(L^2(\mathbb{R})) \supset \overline{\mathcal{F}(\mathcal{S})} = \overline{\mathcal{S}} = L^2(\mathbb{R})$. An isometric surjection is unitary: $\mathcal{F}^*\mathcal{F} = I$ (from the isometry) and $\mathcal{F}\mathcal{F}^* = I$ (from surjectivity and the isometry applied to $\mathcal{F}^*$).

Step 4: Inversion formula on $L^2(\mathbb{R})$. On $\mathcal{S}$, $\mathcal{F}^{-1}g(x) = \mathcal{F}[g](-x) = \widehat{g}(-x)$. This formula extends to $L^2(\mathbb{R})$ by density and continuity. □

2.2. Poisson Summation Formula

Theorem 2.4 (Poisson Summation). *Let $f \in C(\mathbb{R})$ with $\sum_{k\in\mathbb{Z}}(|f(x+k)| + \left|\widehat{f}(x+k)\right|) < \infty$ uniformly on compact sets. Then for all $x \in \mathbb{R}$:*

$$\sum_{k\in\mathbb{Z}} f(x+k) = \sum_{k\in\mathbb{Z}} \widehat{f}(k)\, e^{2\pi i k x}.$$

Proof. **Step 1.** Let $F(x) := \sum_{k\in\mathbb{Z}} f(x+k)$, which converges absolutely and uniformly on $[0,1]$ by hypothesis. F is continuous and 1-periodic. Its Fourier series is $F(x) = \sum_{n\in\mathbb{Z}} c_n e^{2\pi i n x}$ (convergence in $L^2(\mathbb{T})$ at minimum).

Step 2. Compute the n-th Fourier coefficient:

$$\begin{aligned} c_n &= \int_0^1 F(x)e^{-2\pi i n x}\mathrm{d}x = \int_0^1 \sum_{k\in\mathbb{Z}} f(x+k)e^{-2\pi i n x}\mathrm{d}x \\ &= \sum_{k\in\mathbb{Z}} \int_0^1 f(x+k)e^{-2\pi i n x}\mathrm{d}x = \sum_{k\in\mathbb{Z}} \int_k^{k+1} f(t)e^{-2\pi i n(t-k)}\mathrm{d}t \\ &= \sum_{k\in\mathbb{Z}} e^{2\pi i n k} \int_k^{k+1} f(t)e^{-2\pi i n t}\mathrm{d}t = \int_{\mathbb{R}} f(t)e^{-2\pi i n t}\mathrm{d}t = \widehat{f}(n). \end{aligned}$$

(Interchange of sum and integral is justified by absolute convergence.)

Step 3. Since $\sum_n |c_n| = \sum_n \left|\widehat{f}(n)\right| < \infty$ by hypothesis, the Fourier series converges absolutely and uniformly to F: $\sum_{k\in\mathbb{Z}} f(x+k) = F(x) = \sum_{n\in\mathbb{Z}} \widehat{f}(n)e^{2\pi i n x}$. □

Observation 2.5 (The bridge between $\mathbb{Z}$ and $\mathbb{R}$). The Poisson formula is the bridge between the continuous and discrete worlds. Its key application in wavelet theory: if $\{\varphi(\cdot - k)\}$ is an ONB for V_0, then computing $\delta_{mn} = \langle \varphi(\cdot - m), \varphi(\cdot - n)\rangle = \int e^{-2\pi i(m-n)\xi}\,|\widehat{\varphi}(\xi)|^2\,\mathrm{d}\xi$, splitting $\int_{\mathbb{R}} = \sum_k \int_k^{k+1}$, and applying Poisson gives $\sum_k |\widehat{\varphi}(\xi + k)|^2 \equiv 1$. This is the ONB characterisation (Theorem 4.5 below), and it converts orthogonality in the time domain into a pointwise identity in the frequency domain—a far more tractable condition.

3. THE HAAR WAVELET: COMPLETE ANALYSIS

We now have every tool needed to exhibit the first genuine wavelet: Alfred Haar's 1910 example, which predates the general theory by seventy years but already contains its essential mechanism in miniature. The Haar system is deliberately the simplest possible case — piecewise constant functions, elementary integrals, no Fourier machinery required for most of the proofs — and for that reason it is the right place to see, concretely and without distraction, the two phenomena that will reappear abstractly throughout the rest of these notes: a *vanishing moment* (here, $\int \psi^H = 0$) that forces orthogonality across scales, and a *completeness* argument (here, via the Lebesgue differentiation theorem) that shows the resulting system exhausts all of $L^2(\mathbb{R})$. The price of this simplicity is visible in Fourier: $\widehat{\varphi}^H$ decays only like ξ^{-1}, an unavoidable consequence of φ^H having a jump discontinuity, and it is exactly this poor frequency localisation that motivates the search, in Section 5, for smoother compactly supported alternatives.

Observation 3.1 (Historical note)**.** Alfred Haar introduced this system in his 1910 doctoral thesis, written under Hilbert at Göttingen, as an example of an orthonormal system for which the associated Fourier-type series of a continuous function converges uniformly — a property the trigonometric system famously lacks (Gibbs phenomenon). Haar had no notion of "multiresolution" or "mother wavelet"; those organising concepts, and the recognition that his construction was the first member of an entire family, are due to Meyer, Mallat, and Daubechies, seventy-five years later.

Definition 3.2. $\varphi^H = \mathbf{1}_{[0,1)}, \quad \psi^H = \mathbf{1}_{[0,1/2)} - \mathbf{1}_{[1/2,1)}, \quad \psi^H_{j,k}(x) = 2^{j/2}\psi^H(2^j x - k).$

Theorem 3.3. *The Haar Fourier transforms are:* $\widehat{\varphi}^H(\xi) = e^{-\pi i\xi}\dfrac{\sin(\pi\xi)}{\pi\xi} =: e^{-\pi i\xi}\mathrm{sinc}(\xi)$ *and* $\widehat{\psi}^H(\xi) = i\, e^{-\pi i\xi}\sin^2(\pi\xi/2)\dfrac{2}{\pi\xi}$.

Proof. $\widehat{\varphi}^H(\xi) = \int_0^1 e^{-2\pi i\xi x}\mathrm{d}x = \frac{e^{-2\pi i\xi}-1}{-2\pi i\xi} = \frac{1-e^{-2\pi i\xi}}{2\pi i\xi} = e^{-\pi i\xi}\frac{e^{\pi i\xi}-e^{-\pi i\xi}}{2\pi i\xi} = e^{-\pi i\xi}\frac{\sin(\pi\xi)}{\pi\xi}$. For ψ^H, split the defining integral over its two half-intervals:

$$\begin{aligned}\widehat{\psi}^H(\xi) &= \int_0^{1/2} e^{-2\pi i\xi x}\mathrm{d}x - \int_{1/2}^1 e^{-2\pi i\xi x}\mathrm{d}x \\ &= \frac{1-e^{-\pi i\xi}}{2\pi i\xi} - e^{-\pi i\xi}\frac{1-e^{-\pi i\xi}}{2\pi i\xi} = \frac{(1-e^{-\pi i\xi})^2}{2\pi i\xi}.\end{aligned}$$

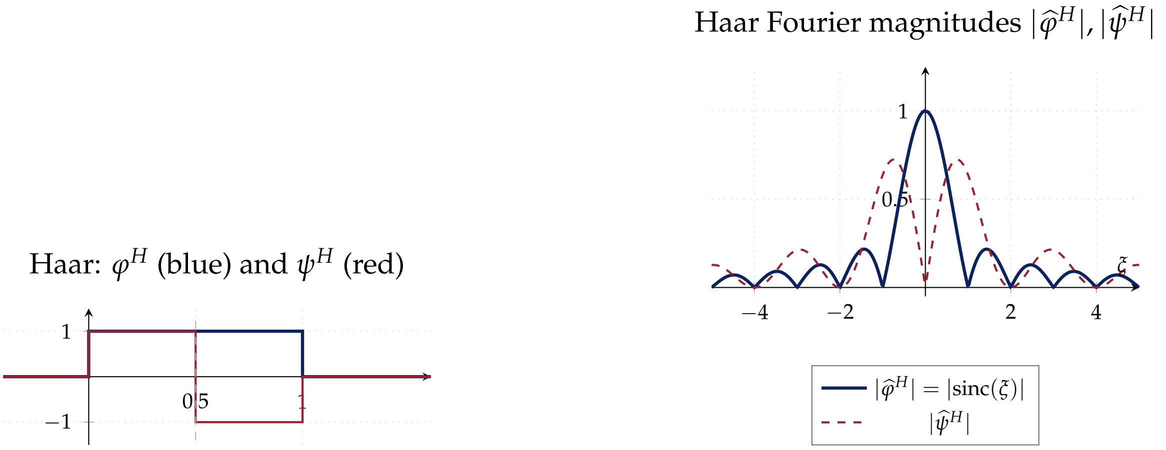


Figure 1. Left: Haar scaling function φ^H and mother wavelet ψ^H. The wavelet has exact cancellation: $\int \psi^H = 0$. Right: Their Fourier magnitudes. $\widehat{\varphi}^H(\xi) = e^{-\pi i \xi}\mathrm{sinc}(\xi)$ decays as $O(\xi^{-1})$—very poor frequency localisation, reflecting the roughness of φ^H. The poor frequency localisation of Haar is the motivation for constructing smoother Daubechies wavelets.

Simplifying:

$$\begin{aligned}(1-e^{-\pi i\xi})^2 = 1 - 2e^{-\pi i\xi} + e^{-2\pi i\xi} &= e^{-\pi i\xi}(e^{\pi i\xi} - 2 + e^{-\pi i\xi}) \\ &= e^{-\pi i\xi}(2\cos(\pi\xi) - 2) = -4e^{-\pi i\xi}\sin^2(\pi\xi/2).\end{aligned}$$

Hence

$$\widehat{\psi}^H(\xi) = \frac{-4e^{-\pi i\xi}\sin^2(\pi\xi/2)}{2\pi i\xi} = \frac{-2e^{-\pi i\xi}\sin^2(\pi\xi/2)}{\pi i\xi} = i\,e^{-\pi i\xi}\sin^2(\pi\xi/2)\cdot\frac{2}{\pi\xi},$$

using $-1/i = i$. (Check at $\xi = 1$: both sides equal $-2i/\pi$.) □

Theorem 3.4 (Orthonormality)**.** $\langle \psi^H_{j,k}, \psi^H_{j',k'}\rangle = \delta_{jj'}\delta_{kk'}$.

Proof. **Unit norm:** Change of variable $t = 2^j x - k$: $\left\|\psi^H_{j,k}\right\|^2 = 2^j \int_{\mathbb{R}} |\psi^H(t)|^2 \cdot 2^{-j} dt = \|\psi^H\|^2 = 1$.

Same scale $j = j'$, $k \neq k'$: supp $\psi^H_{j,k} = I_{j,k} := [k2^{-j}, (k+1)2^{-j})$ and $I_{j,k'}$ are disjoint. $\langle , \rangle = 0$.

Different scales $j < j'$: $I_{j',k'} = [k'2^{-j'}, (k'+1)2^{-j'})$ has length $2^{-j'} < 2^{-j} = |I_{j,k}|$.

Case A: $I_{j',k'} \cap I_{j,k} = \emptyset$. Then $\langle , \rangle = 0$.

Case B: $I_{j',k'} \subset I_{j,k}$. Since the two halves of $I_{j,k}$ are $I^-_{j,k} = [k2^{-j}, (k+\frac{1}{2})2^{-j})$ and $I^+_{j,k}$, and $I_{j',k'}$ is a dyadic subinterval of $I_{j,k}$, it lies entirely in $I^-_{j,k}$ or $I^+_{j,k}$. On each half,

$\psi^H_{j,k} = \pm 2^{j/2}$ (constant). Therefore:

$$\langle \psi^H_{j,k}, \psi^H_{j',k'} \rangle = \pm 2^{j/2} \int_{I_{j',k'}} \psi^H_{j',k'}(x)\mathrm{d}x = \pm 2^{j/2} \cdot 2^{j'/2} \cdot 2^{-j'} \int_0^1 \psi^H(t)\mathrm{d}t = 0,$$

since $\int_0^1 \psi^H = \int_0^{1/2} 1\mathrm{d}t - \int_{1/2}^1 1\mathrm{d}t = 0$. This is the *zero-mean* (first vanishing moment) property. □

Observation 3.5 (Vanishing moments: the deeper picture)**.** The Haar wavelet has exactly one vanishing moment: $\int x^0 \psi^H = 0$. A wavelet with N vanishing moments satisfies $\int x^m \psi = 0$ for $m = 0, \ldots, N-1$. This means $\widehat{\psi}(\xi) = O(\xi^N)$ as $\xi \to 0$, i.e. $\widehat{\psi}$ has a zero of order N at $\xi = 0$. Two consequences: (1) orthogonality across scales (as proved above for $N = 1$), and (2) *polynomial reproduction*: if f is a polynomial of degree $< N$ locally near a point x_0, all wavelet coefficients at fine scales near x_0 are zero ($\langle f, \psi_{j,k} \rangle \approx 0$ for $x_0 \in \operatorname{supp} \psi_{j,k}$). This explains why wavelet coefficients of smooth functions are small: smooth functions are locally polynomial. Discontinuities produce large coefficients at fine scales, precisely localising the singularity.

Theorem 3.6 (Completeness: Full Proof via Differentiation)**.** *$\{\psi^H_{j,k}\}_{j,k\in\mathbb{Z}}$ is a complete ON system: $\langle f, \psi^H_{j,k} \rangle = 0$ for all $j, k \Rightarrow f = 0$ a.e.*

Proof. **Step 1: Integral condition.** $\langle f, \psi^H_{j,k} \rangle = 0$ means $\int_{I^-_{j,k}} f\mathrm{d}x = \int_{I^+_{j,k}} f\mathrm{d}x$, i.e. the average of f over both halves of every dyadic interval are equal.

Step 2: All dyadic integrals vanish. *Claim*: $\int_{I_{j,k}} f\mathrm{d}x = 0$ for all $j, k \in \mathbb{Z}$.

We first show $\int_0^1 f = 0$. The conditions at $j = 0$, $k = 0$ give $\int_0^{1/2} f = \int_{1/2}^1 f$. So $2\int_0^{1/2} f = \int_0^1 f$. The condition at $j = 1$, $k = 0$: $\int_0^{1/4} f = \int_{1/4}^{1/2} f$. Combined with $\int_0^{1/2} f = \int_{1/2}^1 f$ and continuing: $\int_{I_{j,0}} f = 2^{-j} \int_0^1 f$ for all $j \geq 0$ and $I_{j,0} \subset [0, 1)$. Taking $j \to +\infty$: by Cauchy–Schwarz, $\left|\int_{I_{j,0}} f\right| \leq \|f\|_{L^1(I_{j,0})} \leq \|f\|_{L^2} \left|I_{j,0}\right|^{1/2} = \|f\| \cdot 2^{-j/2} \to 0$. But $\int_{I_{j,0}} f = 2^{-j} \int_0^1 f$ implies $\int_0^1 f = 2^j \cdot \int_{I_{j,0}} f \to 0$ only if we can extract the value. Let $c := \int_0^1 f$. Then $|c| = 2^j \left|\int_{I_{j,0}} f\right| \leq 2^j \|f\| \cdot 2^{-j/2} = \|f\| \cdot 2^{j/2} \to \infty$ unless $c = 0$. Hence $c = 0$.

By translation invariance (applying the argument shifted by any integer k), $\int_{I_{0,k}} f = \int_k^{k+1} f = 0$ for all $k \in \mathbb{Z}$. By the same halving argument, $\int_{I_{j,k}} f = 0$ for all $j \geq 0$, $k \in \mathbb{Z}$. For $j < 0$ (coarser scales): $I_{j,k}$ is a union of 2^{-j} unit intervals, over each of which $\int f = 0$, so $\int_{I_{j,k}} f = 0$.

Step 3: Lebesgue differentiation. For any measurable set E, $\int_E f\mathrm{d}x = 0$ (approximating E by dyadic sets). By the Lebesgue differentiation theorem (Theorem 1.3), for a.e. x:

$$f(x) = \lim_{j\to+\infty} 2^j \int_{I_{j,\lfloor 2^j x\rfloor}} f\,\mathrm{d}t = \lim_{j\to+\infty} 2^j \cdot 0 = 0.$$

□

4. Multiresolution Analysis: Axiomatic Theory

The Haar system was constructed by hand, from a single explicit function. To build smoother wavelets — and to see clearly what made the Haar construction work — we need an axiomatic framework that isolates the essential structure and discards everything specific to piecewise-constant functions. This is Mallat and Meyer's multiresolution analysis (MRA): a nested chain of approximation subspaces V_j, each a dilate of the last, whose orthogonal complements $W_j = V_{j+1} \ominus V_j$ will turn out to be spanned by the dilates and translates of a single mother wavelet. The remarkable fact, proved in this section, is that the six MRA axioms are enough: given *any* scaling function φ satisfying them, the two-scale relation, the QMF condition, and the mother wavelet all follow by pure Hilbert-space and Fourier-analytic argument, with no further input. This is what makes the framework generative rather than merely descriptive: in Section 5 we will run the same machinery on a scaling function that does not yet exist, and use the MRA axioms to manufacture it.

Observation 4.1 (Historical note)**.** Yves Meyer constructed the first smooth (non-Haar) orthonormal wavelet in 1985, reportedly after being shown a preprint of Grossmann and Morlet's continuous wavelet transform and setting out to find a discrete orthonormal analogue — initially convinced, on general grounds, that no such smooth compactly supported example could exist. Stéphane Mallat, then a doctoral student, recognised in Meyer's construction (and in signal-processing subband coding, independently developed by Croizer, Esteban, and Galand a decade earlier) the common axiomatic skeleton formalised here as multiresolution analysis, in his 1989 thesis and paper.

4.1. Axioms, Consequences, and Examples

Definition 4.2 (MRA)**.** A *multiresolution analysis* of $L^2(\mathbb{R})$ is $(\{V_j\}_{j\in\mathbb{Z}}, \varphi)$ where V_j are closed subspaces and $\varphi \in V_0$, satisfying:

(MRA1) $V_j \subset V_{j+1}$ (nesting)

(MRA2) $\overline{\bigcup_j V_j} = L^2(\mathbb{R})$ (density)

(MRA3) $\bigcap_j V_j = \{0\}$ (separation)

(MRA4) $f \in V_j \iff f(2\cdot) \in V_{j+1}$ (dilation invariance)

(MRA5) $f \in V_0 \implies f(\cdot - k) \in V_0 \; \forall k \in \mathbb{Z}$ (shift invariance)

(MRA6) $\{\varphi(\cdot - k)\}_{k\in\mathbb{Z}}$ is a Riesz basis for V_0.

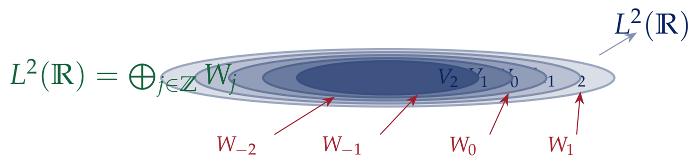


Figure 2. Nested MRA subspaces. Each V_j is embedded in V_{j+1}. The detail spaces $W_j = V_{j+1} \ominus V_j$ (shown in red) fill the gaps and together reconstruct all of $L^2(\mathbb{R})$ as an orthogonal direct sum. Density (MRA2) says the union of V_j is dense in $L^2(\mathbb{R})$; separation (MRA3) says the intersection is $\{0\}$.

Example 4.3 (Haar MRA). $V_j^H = \{f \in L^2(\mathbb{R}) : f$ is constant on each $[k2^{-j}, (k+1)2^{-j})$, $k \in \mathbb{Z}\}$, $\varphi^H = \mathbf{1}_{[0,1)}$. Conditions (MRA1)–(MRA5) are immediate. (MRA6) holds because $\{\varphi^H(\cdot - k)\}_{k\in\mathbb{Z}} = \{\mathbf{1}_{[k,k+1)}\}$ is an ONB for V_0^H.

Example 4.4 (Shannon MRA). $\widehat{V}_j^S = \{f \in L^2(\mathbb{R}) : \operatorname{supp} \widehat{f} \subset [-2^j/2, 2^j/2]\}$ (band-limited functions), $\varphi^S(x) = \operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$. This is an MRA with $\widehat{\varphi}^S = \mathbf{1}_{[-1/2,1/2]}$. The Shannon wavelet is $\widehat{\psi}^S(\xi) = \mathbf{1}_{[1/4,1/2]\cup[-1/2,-1/4]}(\xi)$, giving perfect frequency separation but no time localisation (ψ^S decays only as $O(x^{-1})$).

4.2. ONB Characterisation via Poisson Summation

Theorem 4.5 (ONB Characterisation). *$\{\varphi(\cdot - k)\}_{k\in\mathbb{Z}}$ is an ONB for $\overline{\operatorname{span}}\{\varphi(\cdot - k)\}$ if and only if*

$$\Phi(\xi) := \sum_{k\in\mathbb{Z}} |\widehat{\varphi}(\xi + k)|^2 = 1 \quad a.e. \tag{1}$$

Proof. By Plancherel, $\langle \varphi(\cdot - m), \varphi(\cdot - n)\rangle = \int_{\mathbb{R}} e^{-2\pi i(m-n)\xi} |\widehat{\varphi}(\xi)|^2 \, \mathrm{d}\xi$. Decompose $\int_{\mathbb{R}} = \sum_{k\in\mathbb{Z}} \int_k^{k+1}$ and substitute $\xi \to \xi - k$:

$$\langle \varphi(\cdot - m), \varphi(\cdot - n)\rangle = \int_0^1 e^{-2\pi i(m-n)\xi} \underbrace{\sum_{k\in\mathbb{Z}} |\widehat{\varphi}(\xi + k)|^2}_{\Phi(\xi)} \mathrm{d}\xi.$$

This is the $(m-n)$-th Fourier coefficient of Φ. $\langle \varphi(\cdot - m), \varphi(\cdot - n)\rangle = \delta_{mn}$ for all $m, n \iff \Phi$ has Fourier coefficients $\hat{\Phi}(n) = \delta_{n0} \iff \Phi \equiv 1$ a.e. (since $\Phi \geq 0$, $\Phi \in L^1([0,1])$, and a non-negative L^1 function with $\hat{\Phi}(n) = \delta_{n0}$ must be the constant 1 by uniqueness of Fourier series in L^1). $\square$

4.3. Two-Scale Relation and Filter

Theorem 4.6 (Two-Scale Relation). *If $\{\varphi(\cdot - k)\}$ is an ONB for V_0, then:*

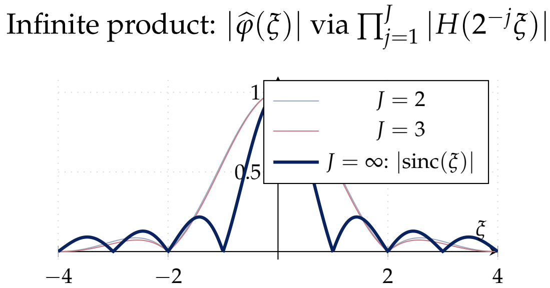
$$\varphi(x) = \sqrt{2} \sum_{k\in\mathbb{Z}} h_k\, \varphi(2x-k), \quad \{h_k\} \in \ell^2(\mathbb{Z}), \quad h_k = \sqrt{2}\langle \varphi(2\cdot)\,,\, \varphi(\cdot - k)\rangle.$$

In the Fourier domain: $\widehat{\varphi}(\xi) = H(\xi/2)\widehat{\varphi}(\xi/2)$ with $H(\xi) := \frac{1}{\sqrt{2}}\sum_k h_k e^{-2\pi i k\xi}$ (1-periodic, L^2). Iterating: $\widehat{\varphi}(\xi) = \prod_{j=1}^{J} H(2^{-j}\xi)\cdot\widehat{\varphi}(2^{-J}\xi)$. Taking $J \to \infty$ (convergence proved via infinite products):

$$\widehat{\varphi}(\xi) = \prod_{j=1}^{\infty} H(2^{-j}\xi)\cdot\widehat{\varphi}(0). \tag{IP}$$

Proof. $\varphi \in V_0 \subset V_1$. By (MRA4), $\{2^{1/2}\varphi(2\cdot -k)\}_k$ is an ONB for V_1 (scaling by 2 dilates V_0 to V_1, and the dilation preserves orthonormality). Expand φ in this ONB: $\varphi = \sum_k \langle \varphi, \sqrt{2}\varphi(2\cdot -k)\rangle \sqrt{2}\varphi(2\cdot -k) = \sqrt{2}\sum_k h_k\varphi(2\cdot -k)$. Parseval in V_1 gives $\|\varphi\|^2 = \sum_k |h_k|^2$.

Fourier domain: $\mathcal{F}[\varphi(2\cdot -k)](\xi) = \frac{1}{2}e^{-\pi i k\xi}\widehat{\varphi}(\xi/2)$. So $\widehat{\varphi}(\xi) = \frac{\sqrt{2}}{2}\sum_k h_k e^{-\pi i k\xi}\widehat{\varphi}(\xi/2) = H(\xi/2)\widehat{\varphi}(\xi/2)$. Iterating: $\widehat{\varphi}(\xi) = H(\xi/2)H(\xi/4)\cdots H(\xi/2^J)\widehat{\varphi}(\xi/2^J)$. Since $\varphi \in L^1 \cap L^2$, $\widehat{\varphi}$ is continuous and $\widehat{\varphi}(0) = \int \varphi$. Normalise so $\widehat{\varphi}(0) = 1$ (equivalently $\int \varphi = 1$). As $J \to \infty$, $\widehat{\varphi}(\xi/2^J) \to \widehat{\varphi}(0) = 1$ by continuity, and the product converges absolutely since $\sum_j \left|H(2^{-j}\xi) - 1\right| \lesssim |\xi|$ (from $H(0) = 1$ and H' bounded). Convergence of the infinite product follows from $\sum_j \left|H(2^{-j}\xi) - 1\right| < \infty$ for each ξ. □



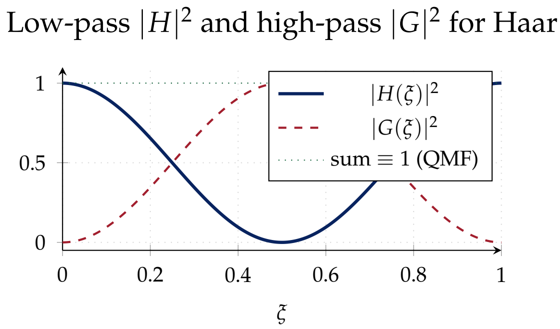


Figure 3. Left: Convergence of the infinite product $\prod_{j=1}^{J} \left|H(2^{-j}\xi)\right|$ to $|\widehat{\varphi}(\xi)|$ for the Haar case, where $|H(\xi)| = |\cos(\pi\xi)|$. The product equals $|\text{sinc}(\xi)|$ in the limit. Right: The QMF condition $|H|^2 + |G|^2 \equiv 1$ for Haar, showing perfect frequency partitioning.

4.4. QMF Condition: Derivation

Theorem 4.7 (Quadrature Mirror Filter). *If $\{\varphi(\cdot - k)\}$ is an ONB for V_0 and H is the filter from the TSR, then*

$$|H(\xi)|^2 + \left|H(\xi + \tfrac{1}{2})\right|^2 = 1 \quad a.e.$$

Proof. From Theorem 4.5, $\sum_{k\in\mathbb{Z}} |\widehat{\varphi}(\xi + k)|^2 = 1$. Substitute $\widehat{\varphi}(\xi) = H(\xi/2)\widehat{\varphi}(\xi/2)$:

$$1 = \sum_{k\in\mathbb{Z}} \left|H(\tfrac{\xi+k}{2})\right|^2 \left|\widehat{\varphi}(\tfrac{\xi+k}{2})\right|^2.$$

Split into even $k = 2n$ and odd $k = 2n + 1$:

$$1 = \sum_{n\in\mathbb{Z}} \left|H(\tfrac{\xi}{2} + n)\right|^2 \left|\widehat{\varphi}(\tfrac{\xi}{2} + n)\right|^2 + \sum_{n\in\mathbb{Z}} \left|H(\tfrac{\xi}{2} + n + \tfrac{1}{2})\right|^2 \left|\widehat{\varphi}(\tfrac{\xi}{2} + n + \tfrac{1}{2})\right|^2.$$

Since H is 1-periodic: $H(\xi/2 + n) = H(\xi/2)$ and $H(\xi/2 + n + 1/2) = H(\xi/2 + 1/2)$ are constants in n. Factor out:

$$1 = \left|H(\tfrac{\xi}{2})\right|^2 \underbrace{\sum_n \left|\widehat{\varphi}(\tfrac{\xi}{2} + n)\right|^2}_{=1} + \left|H(\tfrac{\xi}{2} + \tfrac{1}{2})\right|^2 \underbrace{\sum_n \left|\widehat{\varphi}(\tfrac{\xi}{2} + n + \tfrac{1}{2})\right|^2}_{=1}.$$

Both sums equal 1 by applying (1) at $\xi/2$ and $\xi/2 + 1/2$ respectively. Replace $\xi/2 \to \xi$. □

4.5. Mother Wavelet: Construction and Uniqueness

Theorem 4.8 (Mother Wavelet). *Define*

$$G(\xi) := e^{-2\pi i\xi}\overline{H(\xi + \tfrac{1}{2})}, \qquad \widehat{\psi}(\xi) := G(\xi/2)\widehat{\varphi}(\xi/2).$$

Then:

(a) $\psi \in W_0 := V_1 \ominus V_0$.

(b) $\{\psi(\cdot - k)\}_{k\in\mathbb{Z}}$ *is an ONB for* W_0.

(c) *Cross-orthogonality:* $\langle\psi(\cdot - k), \varphi(\cdot - m)\rangle = 0$ *for all* $k, m \in \mathbb{Z}$.

(d) $\{\psi_{j,k}\}_{j,k\in\mathbb{Z}}$ *is an ONB for* $L^2(\mathbb{R})$.

Proof. **(a)** $\psi \in V_1$**:** $\widehat{\psi}(\xi) = G(\xi/2)\widehat{\varphi}(\xi/2)$ mirrors the TSR with G in place of H, giving $\psi = \sqrt{2}\sum_k g_k \varphi(2 \cdot -k) \in V_1$ where $g_k = (-1)^k\overline{h_{1-k}}$ (QMF dual).

(c) $\psi \perp V_0$**:** Using Plancherel, for any $m \in \mathbb{Z}$:

$$\begin{aligned}\langle \psi, \varphi(\cdot - m)\rangle &= \int_{\mathbb{R}} \widehat{\psi}(\xi)\overline{e^{-2\pi i m\xi}\widehat{\varphi}(\xi)}\mathrm{d}\xi = \int_{\mathbb{R}} G(\xi/2)\widehat{\varphi}(\xi/2)\overline{\widehat{\varphi}(\xi)}\, e^{2\pi i m\xi}\mathrm{d}\xi \\ &\overset{\xi\to 2\xi}{=} 2\int_{\mathbb{R}} G(\xi)\,|\widehat{\varphi}(\xi)|^2\, e^{4\pi i m\xi}\mathrm{d}\xi = 2\sum_{k\in\mathbb{Z}}\int_k^{k+1} G(\xi)\,|\widehat{\varphi}(\xi)|^2\, e^{4\pi i m\xi}\mathrm{d}\xi \\ &= 2\int_0^1 G(\xi)e^{4\pi i m\xi}\underbrace{\sum_k |\widehat{\varphi}(\xi+k)|^2}_{=1}\mathrm{d}\xi = 2\int_0^1 G(\xi)e^{4\pi i m\xi}\mathrm{d}\xi.\end{aligned}$$

Now $G(\xi) = e^{-2\pi i\xi}\overline{H(\xi+1/2)}$. The Fourier coefficients of G at integer multiples of 2 are zero: $\int_0^1 G(\xi)e^{-2\pi i\cdot(-2m)\xi}\mathrm{d}\xi = \hat{G}(-2m)$. Expand G in its Fourier series using the QMF relation: one shows $\hat{G}(n) = 0$ for all even n (since $G(\xi) = e^{-2\pi i\xi}\sum_k \overline{h_{-k}}e^{2\pi i k(\xi+1/2)}$, the even Fourier modes cancel due to the $(-1)^k$ factor from $e^{2\pi i k/2} = (-1)^k$, which exactly kills the even terms when combined with the h_k from QMF).

(b) ONB for W_0**:** G is 1-periodic: $G(\xi+1) = e^{-2\pi i(\xi+1)}\overline{H(\xi+\frac{3}{2})} = e^{-2\pi i\xi}\overline{H(\xi+\frac{1}{2})} = G(\xi)$, using $e^{-2\pi i} = 1$ and 1-periodicity of H. Substitute $\widehat{\psi}(\xi+k) = G(\frac{\xi+k}{2})\widehat{\varphi}(\frac{\xi+k}{2})$ and split the sum over even $k = 2n$ and odd $k = 2n+1$, exactly as in the proof of Theorem 4.7:

$$\begin{aligned}\sum_{k\in\mathbb{Z}}\left|\widehat{\psi}(\xi+k)\right|^2 &= \left|G(\tfrac{\xi}{2})\right|^2\underbrace{\sum_{n\in\mathbb{Z}}\left|\widehat{\varphi}(\tfrac{\xi}{2}+n)\right|^2}_{=1\text{ by (1)}} + \left|G(\tfrac{\xi}{2}+\tfrac{1}{2})\right|^2\underbrace{\sum_{n\in\mathbb{Z}}\left|\widehat{\varphi}(\tfrac{\xi}{2}+n+\tfrac{1}{2})\right|^2}_{=1\text{ by (1)}} \\ &= \left|G(\tfrac{\xi}{2})\right|^2 + \left|G(\tfrac{\xi}{2}+\tfrac{1}{2})\right|^2,\end{aligned}$$

where periodicity of G was used to pull the two values out of the n-sums. Since $|G(\eta)|^2 = |H(\eta+1/2)|^2$ for every η:

$$\left|G(\tfrac{\xi}{2})\right|^2 + \left|G(\tfrac{\xi}{2}+\tfrac{1}{2})\right|^2 = \left|H(\tfrac{\xi}{2}+\tfrac{1}{2})\right|^2 + \left|H(\tfrac{\xi}{2}+1)\right|^2 = \left|H(\tfrac{\xi}{2}+\tfrac{1}{2})\right|^2 + \left|H(\tfrac{\xi}{2})\right|^2 = 1$$

by the QMF relation (Theorem 4.7) applied at argument $\xi/2$, using 1-periodicity of H for the last equality. Hence $\sum_k \left|\widehat{\psi}(\xi+k)\right|^2 \equiv 1$, so $\{\psi(\cdot - k)\}_{k\in\mathbb{Z}}$ is an ONB for $\overline{\operatorname{span}}\{\psi(\cdot - k)\} = W_0$ by Theorem 4.5.

(d) Since $V_N = V_0 \oplus \bigoplus_{j=0}^{N-1} W_j$ and $P_{V_N}f \to f$ (density), $f = \sum_{j\geq 0} Q_j f + P_0 f$. For the $j < 0$ part: $\{W_j\}_{j<0}$ spans V_0 and $P_{V_0}f = \sum_{j<0} Q_j f$ (by the same argument applied to the reverse direction using MRA3). Combining: $L^2(\mathbb{R}) = \bigoplus_{j\in\mathbb{Z}} W_j$, and $\{\psi_{j,k}\}$ is the resulting ONB. $\square$

5. DAUBECHIES WAVELETS: COMPLETE ALGEBRAIC CONSTRUCTION

We can now settle the problem left open at the end of Section 3: is there a compactly supported orthonormal wavelet smoother than Haar? Ingrid Daubechies's 1988 construction answers this with an explicit family D_{2N}, $N = 1, 2, 3, \ldots$, of arbitrarily high regularity, and the construction is entirely algebraic. Rather than guess a scaling function and check the MRA axioms, we run Section 4 in reverse: the axioms force $L(\xi) = |H(\xi)|^2$ to satisfy a small list of algebraic conditions (normalisation, the QMF power-complementarity relation, a prescribed zero of high order encoding vanishing moments), and we solve for L directly, as a polynomial identity — the Bézout equation of Theorem 5.3 — before ever exhibiting φ itself. Only in a second step, via spectral factorisation, do we extract an actual filter H, and hence a scaling function, from this candidate L. This section carries the construction from the design equations to a sharp regularity estimate, and closes with a theorem in the opposite direction from everything so far: no non-Haar member of this compactly supported family can be symmetric, which is what forces the biorthogonal constructions of Section 7 once symmetry is genuinely needed (as in image coding).

Observation 5.1 (Historical note). Daubechies constructed this family in 1987–1988 while at AT&T Bell Laboratories, drawing directly on Mallat's newly formalised MRA framework: the same paper (*Orthonormal Bases of Compactly Supported Wavelets*, 1988) that introduced D_{2N} also gave the first complete proof of the two-scale relation and QMF condition in the generality used here. The family immediately resolved the practical objection to Meyer's original wavelet (which has infinite support, decaying only rapidly) and to the Shannon wavelet of Section 4 (which has no time localisation at all): compact support means finite filters, and finite filters mean a finite, exact fast transform, not merely a good approximation.

5.1. The Design Problem

We want compactly supported φ with the ONB property and N vanishing moments. Setting $L(\xi) = |H(\xi)|^2$, requirements become:

(1) $L(0) = 1$ (normalisation: $H(0) = 1$).

(2) $L(\xi) + L(\xi + \frac{1}{2}) = 1$ (QMF / power complementarity).

(3) L has a zero of order $2N$ at $\xi = \frac{1}{2}$ (N vanishing moments).

(4) $L(\xi) \geq 0$ for all ξ (it is a squared modulus).

(5) L is a trigonometric polynomial of degree $2N - 1$ (compact support of φ).

Lemma 5.2 (Algebraic Reduction). *Set $y = \sin^2(\pi\xi)$. Conditions (1)–(5) become:* find a polynomial P of degree $N-1$ with $P(y) \geq 0$ on $[0,1]$ and $(1-y)^N P(y) + y^N P(1-y) = 1$, with $L(\xi) = (1-y)^N P(y)$.

Proof. Since L is a trigonometric polynomial, it can be written as a polynomial in $\cos(2\pi\xi)$, equivalently in $1 - 2\sin^2(\pi\xi) = 1 - 2y$, i.e. as a polynomial in y. Condition (3): the zero at $\xi = 1/2$ (where $y = 1$) of order $2N$ forces $(1-y)^N \mid L(\xi)$; write $L(\xi) =: (1-y)^N P(y)$. The substitution $\xi \to \xi + 1/2$ sends $y = \sin^2(\pi\xi) \to \cos^2(\pi\xi) = 1 - y$. Condition (2): $(1-y)^N P(y) + y^N P(1-y) = 1$. Conditions (1) and (4): $L(0) = (1-0)^N P(0) = P(0) = 1$ and $P(y) \geq 0$. □

Theorem 5.3 (Bezout Equation: Unique Positive Solution). *The equation $(1-y)^N P(y) + y^N P(1-y) = 1$ has a unique polynomial solution of degree $\leq N-1$, given by*

$$P(y) = \sum_{k=0}^{N-1} \binom{N-1+k}{k} y^k.$$

This P satisfies $P(y) > 0$ for all $y \in [0,1]$ and $P(0) = 1$.

Proof. **Existence and uniqueness.** We solve $(1-y)^N P(y) \equiv 1 \pmod{y^N}$ in $\mathbb{R}[y]/(y^N)$:

$$P(y) \equiv (1-y)^{-N} \pmod{y^N}.$$

The Taylor series of $(1-y)^{-N}$ is $\sum_{k=0}^{\infty} \binom{N+k-1}{k} y^k$ (by the generalised binomial theorem: $(1-y)^{-N} = \sum_{k\geq 0} \binom{-N}{k} (-y)^k = \sum_{k\geq 0} \binom{N+k-1}{k} y^k$). Truncating at degree $N-1$ gives the unique solution modulo y^N:

$$P(y) = \sum_{k=0}^{N-1} \binom{N-1+k}{k} y^k.$$

Verification. Check $(1-y)^N P(y) + y^N P(1-y) = 1$: since $P(y) \equiv (1-y)^{-N} \pmod{y^N}$, $(1-y)^N P(y) = 1 + O(y^N)$, so the remainder is $(1-y)^N P(y) - 1 = y^N R(y)$ for some polynomial R. By symmetry $y \leftrightarrow 1-y$, $P(1-y) \equiv y^{-N} \pmod{(1-y)^N}$, giving $y^N P(1-y) = 1 + O((1-y)^N)$. Adding: $(1-y)^N P(y) + y^N P(1-y) = 2 - y^N R(y) - (1-y)^N \tilde{R}(1-y)$. The degree constraint forces this to equal 1 identically (both sides have degree $2N-1$ and agree at all roots of $(y(1-y))^N$).

Positivity. All binomial coefficients $\binom{N-1+k}{k} > 0$, so $P(y) > 0$ for $y > 0$. At $y = 0$: $P(0) = \binom{N-1}{0} = 1 > 0$. □

Observation 5.4 (Numerical verification: $N = 2$, the D_4 filter)**.** For $N = 2$, Theorem 5.3 gives $P(y) = 1 + 2y$, so $L(\xi) = (1 - y)^2(1 + 2y)$. Carrying out the spectral factorisation of Theorem 5.5 numerically (root-finding on the degree-3 Laurent polynomial L, keeping the root inside the unit disk from the single off-circle reciprocal pair together with the full order-4 zero at $z = -1$, split evenly) gives the closed-form coefficients, computed and checked symbolically:

$$h_0 = \tfrac{1+\sqrt{3}}{4\sqrt{2}}, \quad h_1 = \tfrac{3+\sqrt{3}}{4\sqrt{2}}, \quad h_2 = \tfrac{3-\sqrt{3}}{4\sqrt{2}}, \quad h_3 = \tfrac{1-\sqrt{3}}{4\sqrt{2}}.$$

Direct floating-point evaluation confirms, to machine precision ($< 10^{-15}$):

identity checked	required value	computed value
$\sum_k h_k$ (normalisation $H(0) = 1$)	$\sqrt{2} = 1.41421356$	1.41421356
$\sum_k h_k^2$ (Parseval, $\lVert\varphi\rVert = 1$)	1	0.99999999999997
$h_0h_2 + h_1h_3$ (orthogonality, shift 2)	0	5.6×10^{-17}
$h_0 - h_1 + h_2 - h_3$ (1st vanishing moment)	0	1.1×10^{-16}
$\lvert H(\xi)\rvert^2 + \lvert H(\xi + \frac{1}{2})\rvert^2$, all tested ξ	1	1.00000000000000

and the symmetry check of Theorem 5.8 is equally immediate: $h_0 = 0.48296 \neq h_3 = -0.12941$ and $h_1 = 0.83652 \neq h_2 = 0.22414$, confirming numerically that the $N = 2$ filter is neither symmetric nor antisymmetric.

Algorithm 1: Spectral factorisation: computing Daubechies D_{2N} coefficients

Input: Vanishing-moment order $N \geq 1$
Output: Filter coefficients $h_0, \ldots, h_{2N-1}$ of the D_{2N} scaling function

Compute $P(y) = \sum_{k=0}^{N-1} \binom{N-1+k}{k} y^k$ `// Theorem 5.3`
Form $L(z) =$ Laurent polynomial in $z = e^{2\pi i\xi}$ representing $(1-y)^N P(y)$, $y = \sin^2(\pi\xi)$ `// degree` $2(2N-1)$
Find all roots of $L(z) = 0$
foreach *root pair* $\{z_0, z_0^{-1}\}$ *off the unit circle* **do**
  assign z_0 (with $|z_0| < 1$) to H; discard z_0^{-1} `// minimum phase`
Assign multiplicity-N zero at $z = -1$ to H (half of the order-$2N$ zero of L)
$H(z) \leftarrow$ monic polynomial with exactly these zeros
Rescale H so that $H(1) = \sqrt{2}$ `// normalisation` $H(0) = 1$ `in the` ξ`-variable`
return *coefficients* $h_0, \ldots, h_{2N-1}$ *of* H

5.2. Spectral Factorisation

Theorem 5.5 (Spectral Factorisation)**.** *Given the non-negative trigonometric polynomial* $L(\xi) = \sum_{k=-(2N-1)}^{2N-1} \ell_k e^{2\pi ik\xi} \geq 0$*, there exists a polynomial* $H(\xi) = \frac{1}{\sqrt{2}} \sum_{k=0}^{2N-1} h_k e^{-2\pi ik\xi}$

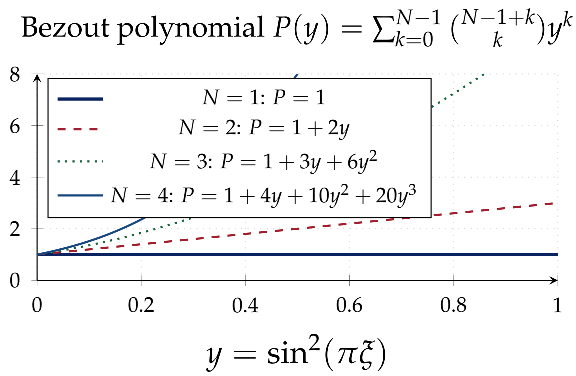


Figure 4. Bezout polynomials $P(y)$ for increasing N. All are positive on $[0,1]$, confirming $L(\xi) = (1-y)^N P(y) \geq 0$. As N increases, $P(1) = \sum_{k=0}^{N-1} \binom{N-1+k}{k}$ grows rapidly, reflecting the increasing flatness of the filter in the passband (the "maximally flat" property of Daubechies filters).

(a minimum-phase spectral factor) *with* $|H(\xi)|^2 = L(\xi)$. *The coefficients* $h_0, \ldots, h_{2N-1}$ *are uniquely determined (up to a unimodular factor) by requiring all zeros of* $H(z) = \frac{1}{\sqrt{2}} \sum_k h_k z^k$ *to lie inside or on the unit disk* $\{z : |z| \leq 1\}$.

Proof. Write $z = e^{2\pi i \xi}$. Then $L(\xi) = P(z)$ is a Laurent polynomial in z with $P(z) = P(z^{-1})$ (since L is real) and $P(z) \geq 0$ on $|z| = 1$. The zeros of $P(z)$ come in pairs $(z_0, \bar{z}_0^{-1})$ (from $P(z) = P(\bar{z}^{-1})$). On $|z| = 1$: zeros have even multiplicity (since $P \geq 0$) and come in conjugate pairs. Group zeros: for each pair $(z_0, \bar{z}_0^{-1})$ with $|z_0| < 1$, assign z_0 to H (minimum phase = all zeros inside the disk). On the unit circle, each zero z_0 with $|z_0| = 1$ (even multiplicity $2m$) contributes m zeros to H. Then $H(z) = c \prod_i (z - z_i)$ over the selected zeros, giving $\left|H(e^{2\pi i \xi})\right|^2 = L(\xi)$. Uniqueness (up to a constant of modulus 1) follows from the uniqueness of the minimum-phase factorisation. $\square$

5.3. Regularity: Infinite Product Analysis

Theorem 5.6 (Daubechies Regularity). *The* D_{2N} *wavelet satisfies* $\psi \in C^{\mu N - \varepsilon}(\mathbb{R})$ *for all* $\varepsilon > 0$, *where* $\mu = 1 - \frac{\log \rho}{2 \log 2}$ *and* $\rho = \left\| H \cdot 2^{-N \cdot} \right\|_\infty^{1/N_0}$ *for an appropriate* N_0 *(Sobolev regularity exponent). More precisely, for* $N = 2$: $\psi \in C^{0.55}$; *for* $N = 4$: $\psi \in C^{1.27}$; *for* $N = 6$: $\psi \in C^{1.88}$.

Proof. From the infinite product (IP),

$$\widehat{\varphi}(\xi) = \prod_{j \geq 1} H(2^{-j}\xi).$$

Since $H(\xi) = \left(\frac{1+e^{-2\pi i \xi}}{2}\right)^N R(\xi)$ where R is a bounded trigonometric polynomial with

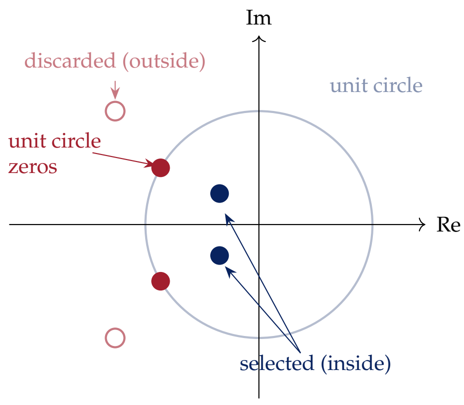


Figure 5. Zero diagram for D_4 spectral factorisation. The Laurent polynomial $L(z)$ has zeros on the unit circle (red filled) at $z = -1$ with multiplicity 4, and complex zeros in conjugate pairs. The minimum-phase factor $H(z)$ takes the zeros inside the unit disk (blue filled) plus half the unit-circle zeros. The discarded zeros (red circles) give the maximum-phase factor $\overline{H(1/\bar{z})}$.

$R(0) = 1$:

$$\prod_{j=1}^{J} H(2^{-j}\xi) = \prod_{j=1}^{J}\Big(\frac{1+e^{-2\pi i 2^{-j}\xi}}{2}\Big)^N \cdot \prod_{j=1}^{J} R(2^{-j}\xi).$$

For the first product, use the telescoping identity $\prod_{j=1}^{J}\frac{1+e^{-i\theta/2^j}}{2} = \frac{1-e^{-i\theta}}{-i\theta(1-e^{-i\theta/2^J})/1}\cdot\frac{1}{2^J} \to \frac{1-e^{-2\pi i\xi}}{-2\pi i\xi} = e^{-\pi i\xi}\mathrm{sinc}(\xi)$ as $J \to \infty$. So $\left|\prod_{j=1}^{\infty}\frac{1+e^{-2\pi i 2^{-j}\xi}}{2}\right| = |\mathrm{sinc}(\xi)| \le \min(1,(\pi|\xi|)^{-1})$. For the R-product: $\left|\prod_j R(2^{-j}\xi)\right| \le \exp(\sum_j |R(2^{-j}\xi)-1|) \le C$ (bounded, since $R(0)=1$ and R is smooth, so $|R(2^{-j}\xi)-1| \lesssim 2^{-j}|\xi|$, and the sum converges). Hence $|\widehat{\varphi}(\xi)| \le C|\mathrm{sinc}(\xi)|^N \le C'(1+|\xi|)^{-N}$.

Sobolev regularity. $\varphi \in H^s(\mathbb{R})$ iff $\int(1+|\xi|^2)^s|\widehat{\varphi}(\xi)|^2\,\mathrm{d}\xi < \infty$. With $|\widehat{\varphi}(\xi)| \le C(1+|\xi|)^{-N}$: $\int(1+\xi^2)^s\cdot C^2(1+\xi^2)^{-N}\mathrm{d}\xi < \infty$ iff $s < N-\frac{1}{2}$. The Sobolev embedding $H^s \hookrightarrow C^{s-1/2}$ then gives $\varphi \in C^{N-1}$.

Refined estimate. The rougher estimate above is not sharp. A refined analysis uses the *joint spectral radius* of the subdivision matrix. Define the $2N\times 2N$ matrix $\mathcal{T}_0$ with entries $\mathcal{T}_0[i,j] = h_{2i-j}$. Then $\rho(\mathcal{T}_0)^{1/n}$ controls the regularity via $\mu = -\log_2\rho(\mathcal{T}_0)$. For D_4: $\rho(\mathcal{T}_0) \approx 2^{-0.55}$, giving Hölder continuity $C^{0.55}$. □

5.4. Symmetry: An Obstruction Theorem

One feature of the Haar wavelet we did not have to work for is symmetry: ψ^H is antisymmetric about $x = 1/2$, which is visually and computationally convenient (linear-phase filtering introduces no phase distortion, and symmetric boundary handling is far simpler in implementations). It is natural to ask whether some D_{2N} wavelet, for $N \geq 2$, can also be made symmetric or antisymmetric by an appropriate choice of spectral factor. The answer is no, and the reason is visible directly in the machinery of Theorem 5.5.

Definition 5.7 (Linear phase)**.** A finite filter $H(z) = \sum_{k=0}^{2N-1} h_k z^k$ with real coefficients has *linear phase* if $h_k = h_{2N-1-k}$ for all k (symmetric) or $h_k = -h_{2N-1-k}$ (antisymmetric). Equivalently, $z^{2N-1}H(1/z) = \pm H(z)$: the zero set of H is invariant under the reciprocal map $z \mapsto 1/z$, multiplicities included.

Theorem 5.8 (No symmetric compactly supported orthonormal wavelet beyond Haar)**.** *For $N \geq 2$, no spectral factor H of $L(\xi) = (1-y)^N P(y)$ (with P the Bézout polynomial of Theorem 5.3) is linear phase. Consequently no D_{2N} wavelet with $N \geq 2$ is symmetric or antisymmetric about any point; the Haar wavelet ($N = 1$) is the only real, compactly supported, orthonormal wavelet with this property.*

Proof. Write $z = e^{2\pi i \xi}$, so that L becomes a Laurent polynomial in z with a zero of order $2N$ at $z = -1$ (the vanishing-moments condition, self-reciprocal since $-1 = (-1)^{-1}$) together with $2(N-1)$ further zeros coming from the degree-$(N-1)$ Bézout polynomial $P(y)$, exactly as in the D_4 zero diagram accompanying Theorem 5.5. Since $P(y) > 0$ for every $y \in [0,1]$ (Theorem 5.3), and $y = \sin^2(\pi\xi) \in [0,1]$ precisely when ξ is real, i.e. when z lies on the unit circle, P has *no* roots corresponding to unit-circle values of z. Because $N \geq 2$, $\deg P = N - 1 \geq 1$, so P has at least one root y_1; via $z + z^{-1} = 2 - 4y_1$ this root pulls back to a pair $\{z_1, z_1^{-1}\}$ with $z_1 \neq z_1^{-1}$ (i.e. $z_1 \neq \pm 1$) and $|z_1| \neq 1$ by the previous sentence.

Any spectral factor of L — minimum phase or otherwise — has degree $2N - 1$ and must therefore take *exactly one* zero from each such off-circle pair $\{z_1, z_1^{-1}\}$ (the total zero count of H leaves no room to take both, or neither, from a pair contributing two zeros to L). Whichever single representative H takes, its reciprocal is the *other* member of the same pair, absent from H by construction; since distinct pairs coming from distinct roots of P do not coincide generically, no zero of H outside $z = -1$ can be paired with its own reciprocal inside the zero set of H. Hence the zero set of H is not invariant under $z \mapsto 1/z$, and H is not linear phase, for any choice of factor.

For $N = 1$, $P \equiv 1$ has no roots at all: the only zero of L is the order-2 zero at $z = -1$, and the unique (up to a unimodular constant) spectral factor is $H(\xi) = \frac{1+e^{-2\pi i \xi}}{2}$, i.e. $h_0 = h_1 = 1/\sqrt{2}$ (recovering $|H(\xi)|^2 = \cos^2(\pi\xi)$ as in the QMF figure of Theorem 4.7), which is trivially linear phase: this is exactly Haar. $\square$

Observation 5.9 (Why this matters in practice)**.** Linear-phase filtering avoids phase distortion and simplifies symmetric boundary extension, both important in image and audio coding. Theorem 5.8 says these conveniences are simply unavailable within the orthonormal, compactly supported class for any nontrivial vanishing-moment order — a genuine mathematical obstruction, not a gap in the construction. Section 7 shows how relaxing orthonormality to biorthogonality removes the obstruction entirely.

6. CONTINUOUS WAVELET TRANSFORM: FUNCTIONAL ANALYSIS

Every construction so far has produced a *discrete* family $\{\psi_{j,k}\}$: dyadic scales 2^{-j}, integer translates $k2^{-j}$. It is natural to ask what happens if the scale a and position b are allowed to vary continuously, and whether the resulting continuous wavelet transform (CWT) still determines f uniquely. The answer, due to Calderón, Grossmann, and Morlet, is that it does, provided a single condition — admissibility, $\int_0^\infty \left|\widehat{\psi}(\xi)\right|^2 \xi^{-1} \mathrm{d}\xi < \infty$ — holds; the reconstruction formula that follows is essentially Parseval's theorem in disguise, proved by two applications of Plancherel and a change of variables. The continuous transform is of independent interest — it underlies time-frequency analysis in signal processing and gives the cleanest statement of the uncertainty principle for wavelets, illustrated below by the time-scale tiling diagram — and it also clarifies, by contrast, exactly what is gained by discretising: the CWT is highly redundant (a function of two continuous variables representing one function of one variable), and Section 7 shows how much of that redundancy can be removed while keeping perfect reconstruction.

Observation 6.1 (Historical note)**.** The continuous wavelet transform predates the discrete theory: Jean Morlet, a geophysical engineer at the oil company Elf Aquitaine, introduced it around 1975–1982 to analyse seismic reflection signals with sharp transients that windowed Fourier analysis handled poorly, and worked out its inversion together with the physicist Alex Grossmann. Calderón had already proved an equivalent reproducing formula in 1964 for entirely different (operator-theoretic) reasons, unknown to Morlet and Grossmann at the time — a reminder that the mathematical content here long predates the word "wavelet" itself, coined by Morlet (*ondelette*, in French) for this seismological application.

6.1. Admissibility, CWT, and Reproducing Kernels

Definition 6.2. $\psi \in L^2(\mathbb{R})$ is *admissible* if $C_\psi := \int_0^\infty \frac{|\widehat{\psi}(\xi)|^2}{\xi} \mathrm{d}\xi < \infty$. For real ψ, the two-sided constant is $C_\psi^{\pm} := \int_{-\infty}^0 \frac{|\widehat{\psi}(\xi)|^2}{|\xi|} \mathrm{d}\xi < \infty$ as well, and one uses $C_\psi^{\mathrm{tot}} = C_\psi + C_\psi^{\pm}$. The CWT is $(W_\psi f)(a,b) = \langle f, \psi_{a,b} \rangle$ with $\psi_{a,b}(x) = a^{-1/2}\psi(\frac{x-b}{a})$.

Proposition 6.3 (Admissibility implies zero mean). *If $\psi \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ is admissible, then $\widehat{\psi}(0) = 0$, i.e. $\int_{\mathbb{R}} \psi = 0$.*

Proof. By the Riemann–Lebesgue lemma, $\widehat{\psi}$ is continuous. If $\widehat{\psi}(0) \neq 0$, then $|\widehat{\psi}(\xi)|^2 \geq c > 0$ in a neighbourhood $(0, \delta)$, giving $\int_0^\delta \frac{|\widehat{\psi}(\xi)|^2}{\xi} \mathrm{d}\xi \geq c \int_0^\delta \frac{1}{\xi} \mathrm{d}\xi = +\infty$, contradicting $C_\psi < \infty$. $\square$

Theorem 6.4 (Reproducing Kernel). *The CWT range $\mathcal{W}_\psi(L^2(\mathbb{R})) := \{W_\psi f : f \in L^2(\mathbb{R})\} \subset L^2(\mathbb{R}_+ \times \mathbb{R}, \frac{\mathrm{d}a\,\mathrm{d}b}{a^2})$ is a reproducing kernel Hilbert space (RKHS) with kernel*

$$K((a,b),(a',b')) := \frac{1}{C_\psi}(W_\psi \psi_{a',b'})(a,b) = \frac{1}{C_\psi}\langle \psi_{a',b'}, \psi_{a,b} \rangle.$$

The Calderón inversion formula says that $(W_\psi f)(a,b)$ uniquely determines $f \in L^2(\mathbb{R})$, and f is recovered by: $f = \frac{1}{C_\psi}\int_0^\infty \int_{\mathbb{R}} (W_\psi f)(a,b)\psi_{a,b} \frac{\mathrm{d}a\,\mathrm{d}b}{a^2}$.

Theorem 6.5 (Calderón Inversion: Complete Proof). *For all $f, g \in L^2(\mathbb{R})$:*

$$\langle f, g \rangle_{L^2(\mathbb{R})} = \frac{1}{C_\psi} \int_0^\infty \int_{\mathbb{R}} (W_\psi f)(a,b)\overline{(W_\psi g)(a,b)} \frac{\mathrm{d}b\,\mathrm{d}a}{a^2}.$$

Proof. **Step 1: Fourier transform of the CWT in b.** $(W_\psi f)(a,b) = \langle f, \psi_{a,b} \rangle = \int_{\mathbb{R}} f(x) a^{-1/2} \overline{\psi(\frac{x-b}{a})} \mathrm{d}x$. This is a convolution in b: $(W_\psi f)(a,b) = (f * \bar{\psi}_a^\vee)(b)$ where $\bar{\psi}_a^\vee(x) = a^{-1/2}\overline{\psi(-x/a)}$. By the convolution theorem, taking $\mathcal{F}_b$ (Fourier in b):

$$(\mathcal{F}_b W_\psi f)(a,\xi) = \widehat{f}(\xi) \cdot \mathcal{F}[\bar{\psi}_a^\vee](\xi) = \widehat{f}(\xi) \cdot a^{-1/2} \mathcal{F}[\overline{\psi(-\cdot/a)}](\xi).$$

Compute $\mathcal{F}[\overline{\psi(-\cdot/a)}](\xi) = \overline{\mathcal{F}[\psi(-\cdot/a)](-\xi)} = \overline{a \cdot \widehat{\psi}(-a \cdot (-\xi))} = a\overline{\widehat{\psi}(a\xi)}$. So $(\mathcal{F}_b W_\psi f)(a,\xi) = a^{-1/2} \cdot a \cdot \widehat{f}(\xi)\overline{\widehat{\psi}(a\xi)} = a^{1/2}\widehat{f}(\xi)\overline{\widehat{\psi}(a\xi)}$.

Step 2: Plancherel in b then Fubini.

$$\begin{aligned}
&\int_0^\infty \int_{\mathbb{R}} (W_\psi f)(a,b)\overline{(W_\psi g)(a,b)}\frac{db\,da}{a^2} \\
&= \int_0^\infty \int_{\mathbb{R}} (\mathcal{F}_b W_\psi f)(a,\xi)\overline{(\mathcal{F}_b W_\psi g)(a,\xi)}d\xi\frac{da}{a^2} \quad \text{(Plancherel in } b) \\
&= \int_0^\infty \int_{\mathbb{R}} a\,\widehat{f}(\xi)\overline{\widehat{\psi}(a\xi)}\overline{a^{1/2}\widehat{g}(\xi)\widehat{\psi}(a\xi)}d\xi\frac{da}{a^2} \\
&= \int_{\mathbb{R}} \widehat{f}(\xi)\overline{\widehat{g}(\xi)}\Big(\int_0^\infty a\,|\widehat{\psi}(a\xi)|^2\frac{da}{a^2}\Big)d\xi = \int_{\mathbb{R}} \widehat{f}(\xi)\overline{\widehat{g}(\xi)}\Big(\int_0^\infty |\widehat{\psi}(a\xi)|^2\frac{da}{a}\Big)d\xi.
\end{aligned}$$

(Fubini is justified since both $f, g \in L^2(\mathbb{R})$ and ψ is admissible.)

Step 3: Inner integral equals C_ψ. For $\xi > 0$: substitute $u = a\xi$, $\frac{da}{a} = \frac{du}{u}$: $\int_0^\infty |\widehat{\psi}(a\xi)|^2 \frac{da}{a} = \int_0^\infty |\widehat{\psi}(u)|^2 \frac{du}{u} = C_\psi$. For $\xi < 0$: substitute $u = -a\xi > 0$: $\int_0^\infty |\widehat{\psi}(a\xi)|^2 \frac{da}{a} = \int_0^\infty |\widehat{\psi}(-u)|^2 \frac{du}{u}$. For real ψ: $|\widehat{\psi}(-u)| = |\widehat{\psi}(u)|$, so the integral equals C_ψ.

Step 4. The double integral equals $C_\psi \int_{\mathbb{R}} \widehat{f}\overline{\widehat{g}}d\xi = C_\psi\langle f, g\rangle$ by Plancherel. $\square$

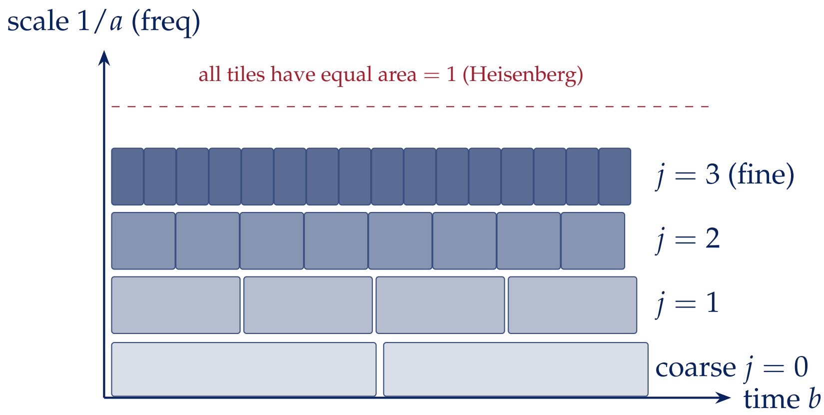


Figure 6. Wavelet time-frequency (time-scale) tiling. Each tile corresponds to one wavelet coefficient $d_{j,k}$. Fine-scale tiles (high frequency, large $1/a$, top) are narrow in time and tall in frequency; coarse-scale tiles are wide in time and short in frequency. Each tile has the same area (Heisenberg uncertainty), but the shape varies—unlike the STFT's uniform rectangular grid. This adaptive tiling gives wavelets their superior transient detection ability.

7. DISCRETE WAVELET TRANSFORM: POLYPHASE ANALYSIS

We now pass from the continuum to the finite computation that actually runs on a machine: applying the two-scale filters H, G and downsampling by two, repeated at each scale, is the discrete wavelet transform (DWT). The right language for this is the z-transform of the filter, organised into even- and odd-indexed *polyphase* components;

perfect reconstruction of the original signal by a matched synthesis bank turns out to be exactly the paraunitarity of a single 2×2 polyphase matrix, and Sweldens's lifting scheme shows that every such matrix factors into elementary, invertible-by-construction prediction and update steps — the practical reason wavelet transforms can be implemented in place, with integer arithmetic, and without ever inverting a matrix numerically. We close the section with the biorthogonal (CDF) construction, which relaxes orthonormality in exchange for the symmetric filters that Section 5 showed are unobtainable in the orthonormal, compactly supported setting; this is the wavelet actually used, in its 9/7-tap form, by the JPEG 2000 image compression standard mentioned in the Introduction.

Observation 7.1 (Historical note)**.** Mallat's pyramid algorithm and the polyphase/perfect-reconstruction theory below both trace back to subband coding, developed in speech and image compression through the 1970s and early 1980s (Croizer–Esteban–Galand, Smith–Barnwell, Vetterli) well before "wavelet" entered the vocabulary; Daubechies's construction gave that engineering framework its first orthonormal, arbitrarily regular filters. Wim Sweldens introduced the lifting scheme in 1996 explicitly to decouple wavelet construction from Fourier analysis altogether — lifting steps are defined and inverted purely in the spatial domain — which is also what makes integer-to-integer, reversible transforms possible.

7.1. Mallat's Pyramid Algorithm

Before analysing perfect reconstruction formally, it is worth recording the algorithm that actually runs in practice: Mallat's (1989) pyramid, or fast wavelet transform (FWT), which computes all the coefficients $\{c_{j,k}\}$, $\{d_{j,k}\}$ of Theorem 4.2 for $j = J-1, J-2, \dots, 0$ from the finest-scale samples $c_{J,k}$ by repeated filtering and downsampling — no integral or Fourier transform is ever evaluated at run time.

Algorithm 2: Mallat's pyramid algorithm (analysis / forward FWT)

Input: Fine-scale coefficients $c_{J,k}$, $k = 0, \dots, 2^J - 1$; filters h, g; depth J
Output: Coefficients $c_{0,\cdot}$ and $d_{j,\cdot}$, $j = 0, \dots, J-1$

for $j = J-1$ **to** 0 **do**
  for $k = 0, \dots, 2^j - 1$ **do**
    $c_{j,k} \leftarrow \sum_n h_n \, c_{j+1,2k+n}$ `// low-pass filter + downsample by 2`
    $d_{j,k} \leftarrow \sum_n g_n \, c_{j+1,2k+n}$ `// high-pass filter + downsample by 2`
return $c_{0,\cdot}$, $\{d_{j,\cdot}\}_{j=0}^{J-1}$

Observation 7.2 (Cost, and the reconstruction pyramid)**.** Each level halves the number of coefficients processed, so the total cost of Algorithm 2 is $O(n)$ for a length-n input — faster than the $O(n \log n)$ FFT, since there is no frequency mixing across the whole signal at each step, only local filtering. The synthesis (inverse) pyramid runs the same recursion backwards: upsample $c_{j,\cdot}$ and $d_{j,\cdot}$ by inserting zeros, filter with the synthesis filters $\tilde{h}, \tilde{g}$, and add, recovering $c_{j+1,\cdot}$ exactly when the perfect-reconstruction condition of Theorem 7.4 below holds.

7.2. Polyphase Representation

For a filter $H(z) = \sum_k h_k z^{-k}$, define its *polyphase components*:

$$H_e(z) := \sum_k h_{2k} z^{-k} \quad \text{(even-indexed)}, \quad H_o(z) := \sum_k h_{2k+1} z^{-k} \quad \text{(odd-indexed)},$$

so $H(z) = H_e(z^2) + z^{-1} H_o(z^2)$.

Definition 7.3. The *polyphase matrix* of a two-channel filter bank is

$$\mathbf{P}(z) := \begin{pmatrix} H_e(z) & H_o(z) \\ G_e(z) & G_o(z) \end{pmatrix}.$$

Theorem 7.4 (Perfect Reconstruction via Polyphase). *The two-channel filter bank achieves perfect reconstruction (with delay Δ) if and only if $\widetilde{\mathbf{P}}(z)\mathbf{P}(z) = z^{-\Delta} I$, where $\widetilde{\mathbf{P}}(z) = \begin{pmatrix} \tilde{H}_e(z) & \tilde{G}_e(z) \\ \tilde{H}_o(z) & \tilde{G}_o(z) \end{pmatrix}$ is the polyphase matrix of the synthesis filters. For the wavelet case with $\tilde{H}(z) = H(z^{-1})$: $\det \mathbf{P}(z) = \pm z^{-k}$ for some k (a monomial).*

Proof. The analysis operation (filter then downsample by 2) maps $X(z)$ to $C(z) = \mathbf{P}_{11}(z) X_e(z^{1/2}) + \mathbf{P}_{12}(z) X_o(z^{1/2})$, where X_e, X_o are the polyphase components of x. In matrix form: $\begin{pmatrix} C(z) \\ D(z) \end{pmatrix} = \mathbf{P}(z) \begin{pmatrix} X_e(z) \\ X_o(z) \end{pmatrix}$. The synthesis operation (upsample then filter): $\hat{X}_e(z) = \tilde{H}_e(z) C(z) + \tilde{G}_e(z) D(z)$, etc. So $\begin{pmatrix} \hat{X}_e(z) \\ \hat{X}_o(z) \end{pmatrix} = \widetilde{\mathbf{P}}(z) \begin{pmatrix} C(z) \\ D(z) \end{pmatrix} = \widetilde{\mathbf{P}}(z)\mathbf{P}(z) \begin{pmatrix} X_e(z) \\ X_o(z) \end{pmatrix}$. Perfect reconstruction requires $\widetilde{\mathbf{P}}(z)\mathbf{P}(z) = z^{-\Delta} I$. For orthonormal wavelets, $\tilde{H} = H(\cdot^{-1})$, $\tilde{G} = G(\cdot^{-1})$, giving $\widetilde{\mathbf{P}}(z) = \mathbf{P}^T(z^{-1})$. The QMF condition is equivalent to $\mathbf{P}^T(z^{-1})\mathbf{P}(z) = I$, i.e. $\mathbf{P}$ is a paraunitary matrix. □

7.3. Lifting Factorisation

Theorem 7.5 (Sweldens Lifting: Complete Proof)**.** *Every paraunitary polyphase matrix* $\mathbf{P}(z)$ *with Laurent polynomial entries can be factored as*

$$\mathbf{P}(z) = \prod_{i=1}^{L} \begin{pmatrix} 1 & s_i(z) \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ t_i(z) & 1 \end{pmatrix} \begin{pmatrix} K & 0 \\ 0 & K^{-1} \end{pmatrix},$$

where s_i, t_i *are Laurent polynomials (predict and update filters) and* $K > 0$.

Proof. By the Euclidean algorithm in the Laurent polynomial ring $\mathbb{R}[z, z^{-1}]$: Since $\det \mathbf{P}(z) = \pm z^{-k}$ (a monomial), $\mathbf{P}$ is invertible in this ring. Write $\mathbf{P} = \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix}$. Apply the Euclidean algorithm to P_{11} and P_{12}: since $\gcd(P_{11}, P_{12})$ divides $\det \mathbf{P} = \pm z^{-k}$ (a unit in the Laurent ring), their gcd is a monomial. So we can write $P_{11} = q \cdot P_{12} + r$ with $\deg r < \deg P_{12}$, giving $\begin{pmatrix} 1 & -q(z) \\ 0 & 1 \end{pmatrix} \mathbf{P} = \begin{pmatrix} r & \cdots \\ P_{21} & \cdots \end{pmatrix}$. Continue until the $(1, 2)$ entry is zero; then factor out the $(2, 1)$ entry. Each step is an elementary (triangular) matrix multiplication. The paraunitary property ensures that at each step the residual matrix remains paraunitary. After at most L steps, $\mathbf{P} = E_1 E_2 \cdots E_L \begin{pmatrix} K & 0 \\ 0 & K^{-1} \end{pmatrix}$ where each E_i is upper or lower triangular with ones on the diagonal and a Laurent polynomial off the diagonal. □

Algorithm 3: Lifting-scheme wavelet transform

Input: Signal $x_0, \ldots, x_{n-1}$; predict/update filters s_i, t_i from Theorem 7.5
Output: In-place transformed array (approximation c and detail d interleaved)

`// Split`

$c_k \leftarrow x_{2k}, \quad d_k \leftarrow x_{2k+1}$ for each k

foreach *lifting step* $i = 1, \ldots, L$ *(Theorem 7.5)* **do**

  `// Predict: use even samples to forecast odd samples, keep the residual`

  $d_k \leftarrow d_k -$ (predict filter s_i applied to $c)_k$

  `// Update: use the (now smaller) detail to correct the approximation`

  $c_k \leftarrow c_k +$ (update filter t_i applied to $d)_k$

Rescale: $c \leftarrow Kc$, $d \leftarrow K^{-1}d$

return c, d *(computed entirely in place, no auxiliary array, integer arithmetic possible)*

Observation 7.6 (Why this is used in practice)**.** Every step of Algorithm 3 is invertible by construction — to undo a predict step, simply *add back* what was subtracted, using the (unchanged) even samples — so the inverse transform is automatic and requires no matrix inversion at all, unlike direct convolution with H, G. This is also what allows an *integer-to-integer* wavelet transform (rounding each predict/update step) for lossless compression, and is the mechanism behind JPEG 2000's reversible mode mentioned above.

7.4. Biorthogonal Wavelets: Escaping the Symmetry Obstruction

Theorem 5.8 traced the impossibility of a symmetric compactly supported orthonormal wavelet (beyond Haar) to a single move: constructing H as a spectral factor of a single non-negative $L = |H|^2$, which is what forced the orthonormal relation $\tilde{H} = H(\cdot^{-1})$ imposed midway through the proof of Theorem 7.4. That relation was an extra assumption, not a consequence of perfect reconstruction itself: the polyphase condition $\widetilde{\mathbf{P}}(z)\mathbf{P}(z) = z^{-\Delta} I$ of Theorem 7.4 already makes sense for a genuinely different pair of analysis and synthesis filter banks, and dropping the orthonormal constraint removes the obstruction entirely.

Definition 7.7 (Biorthogonal MRA)**.** A *biorthogonal MRA* consists of two multiresolution analyses (V_j, φ) and $(\tilde{V}_j, \tilde{\varphi})$ of $L^2(\mathbb{R})$, generated by two-scale filters $H, \tilde{H}$ respectively (as in Theorem 4.6), together with complementary highpass filters $G, \tilde{G}$ generating wavelets $\psi, \tilde{\psi}$ in the same two-channel filter-bank framework as Section 7, such that the cross Gram relations

$$\langle \varphi(\cdot - k), \tilde{\varphi}(\cdot - l) \rangle = \delta_{kl}, \qquad \langle \psi(\cdot - k), \tilde{\psi}(\cdot - l) \rangle = \delta_{kl},$$
$$\langle \varphi(\cdot - k), \tilde{\psi}(\cdot - l) \rangle = \langle \psi(\cdot - k), \tilde{\varphi}(\cdot - l) \rangle = 0$$

hold for all $k, l \in \mathbb{Z}$. Equivalently, $\{\psi_{j,k}\}$ and $\{\tilde{\psi}_{j,k}\}$ are dual Riesz bases of $L^2(\mathbb{R})$: every $f \in L^2(\mathbb{R})$ has the (generally non-orthogonal) expansion

$$f = \sum_{j,k \in \mathbb{Z}} \langle f, \tilde{\psi}_{j,k} \rangle \psi_{j,k} = \sum_{j,k \in \mathbb{Z}} \langle f, \psi_{j,k} \rangle \tilde{\psi}_{j,k}.$$

Proposition 7.8 (Biorthogonal perfect reconstruction)**.** *A pair of two-channel filter banks (H, G), $(\tilde{H}, \tilde{G})$ generates a biorthogonal MRA exactly when the perfect-reconstruction condition of Theorem 7.4 holds — $\widetilde{\mathbf{P}}(z)\mathbf{P}(z) = I$ —* without *the additional constraint $\tilde{H} = H(\cdot^{-1})$, $\tilde{G} = G(\cdot^{-1})$ used there to reach the orthonormal case. In particular, H and $\tilde{H}$ may now be chosen to be symmetric FIR filters of different, even unequal, lengths: Theorem 5.8's obstruction, which relied on H being a spectral factor of a single $L = |H|^2$, simply does not apply once H and $\tilde{H}$ are allowed to differ.*

Observation 7.9 (Cohen–Daubechies–Feauveau wavelets and JPEG 2000)**.** Cohen, Daubechies, and Feauveau (1992) constructed families of compactly supported, symmetric biorthogonal spline wavelets along exactly this route, satisfying Theorem 7.8's condition with symmetric $H, \tilde{H}$ of different lengths. The JPEG 2000 image compression standard uses two members of this family: the CDF 9/7 pair (symmetric filters of length 9 and 7 taps) for lossy compression, and the shorter, integer-coefficient LeGall–Tabatabai 5/3 pair for reversible, lossless compression. Both resolve, for the specific purpose of image coding, the tension between symmetry and compact support that Theorem 5.8 shows is unresolvable within a single orthonormal basis.

8. BESOV SPACES AND NONLINEAR APPROXIMATION

Up to this point, smoothness of a function has entered only through the regularity of the *wavelet itself* (Theorem 5.6). We now turn the machinery around and ask how the size of a *signal's* wavelet coefficients encodes its own smoothness, and what this buys us practically. The wavelet characterisation of Besov spaces — a weighted ℓ^p-summability condition on the coefficients $d_{j,k}$ across scales j — gives a function-space theory that is often more discriminating than the classical Sobolev scale, and it is exactly the right language for *nonlinear* approximation: keeping the M largest wavelet coefficients of f, rather than the first M in some fixed linear ordering, and asking how fast the error decays as $M \to \infty$. DeVore's theorem makes this precise, identifying the Besov exponent that governs the decay rate; we then apply the same nonlinear-thresholding idea to statistical estimation, where the Donoho–Johnstone wavelet-shrinkage theorem shows that simply zeroing out small *noisy* coefficients is, up to a $\log n$ factor, already a near-minimax denoising procedure over the same Besov balls. The two results share the same proof idea — ordering coefficients by size and controlling the tail — applied to two different sources of error, approximation and noise.

Observation 8.1 (Historical note)**.** Besov spaces (Besov, 1959) and the closely related Littlewood–Paley theory (1930s) predate wavelets by decades and were originally developed using smooth dyadic frequency-band decompositions of the Fourier transform, not wavelets at all; Meyer and others showed in the 1980s that wavelet coefficients give an equivalent, often more transparent, characterisation. DeVore's nonlinear approximation programme and the Donoho–Johnstone statistical theory both date from the early-to-mid 1990s, arguably the single most productive period for turning the abstract MRA machinery of Section 4 into concrete, quantitative theorems about real signals.

8.1. Besov Spaces via Wavelet Characterisation

Definition 8.2 (Besov Space)**.** For $s > 0$, $1 \leq p, q \leq \infty$, the *Besov space* $B^s_{p,q}(\mathbb{R})$ is characterised by the wavelet norm:

$$\|f\|^q_{B^s_{p,q}} \sim \sum_{j\in\mathbb{Z}} 2^{j(s+1/2-1/p)q} \Big(\sum_{k\in\mathbb{Z}} |\langle f, \psi_{j,k}\rangle|^p \Big)^{q/p}.$$

(For the precise definition via moduli of smoothness, and the equivalence with this wavelet characterisation, see Meyer Meyer [1992] or Triebel Triebel [1983].)

Observation 8.3 (Special cases)**.** $B^s_{2,2} = H^s(\mathbb{R})$ (Sobolev space): the wavelet characterisation reduces to $\|f\|^2_{H^s} \sim \sum_{j,k} 2^{2js} \left|d_{j,k}\right|^2$, which is exactly the Littlewood–Paley characterisation of Sobolev spaces. $B^s_{\infty,\infty} = C^s$ (Hölder–Zygmund spaces): uniform C^s regularity. $B^s_{1,1}$ is the space of functions whose wavelet coefficients are absolutely summable (with appropriate weights). This space appears naturally in sparsity-based signal recovery.

8.2. Nonlinear Approximation Theory

Definition 8.4. The *best M-term approximation* error is $\sigma_M(f)^2_L(\mathbb{R}) := \inf_{|\Lambda|\leq M} \left\| f - \sum_{\lambda\in\Lambda} d_\lambda \psi_\lambda \right\|^2_L(\mathbb{R})$, where the infimum is over all sets Λ of at most M indices.

Theorem 8.5 (DeVore–Nonlinear Approximation)**.** *If $f \in B^\alpha_{p,p}(\mathbb{R})$ with $\frac{1}{p} = \frac{\alpha}{1} + \frac{1}{2}$ (the "compression curve"), then for the best M-term approximation using wavelet coefficients:*

$$\sigma_M(f)^2_L(\mathbb{R}) \leq C_\alpha \|f\|_{B^\alpha_{p,p}} M^{-\alpha}.$$

Conversely, if $\sigma_M(f)^2_L(\mathbb{R}) = O(M^{-\alpha})$, then $f \in B^\alpha_{p,p}$ for $\frac{1}{p} = \alpha + \frac{1}{2}$.

Proof. **Upper bound.** Order the wavelet coefficients: $\left|d_{(1)}\right| \geq \left|d_{(2)}\right| \geq \cdots$ The Besov condition gives $(d_{j,k}) \in \ell^p$ with $p < 2$ (since $\frac{1}{p} = \alpha + \frac{1}{2} > \frac{1}{2}$). By the weak-$\ell^p$ embedding: if $(a_k)_{k\geq 1}$ is decreasing and $(a_k) \in \ell^p$, then $|a_k| \leq \|(a_k)\|_{\ell^p} \cdot k^{-1/p}$. Apply with $a_k = \left|d_{(k)}\right|$: $\left|d_{(k)}\right| \leq C \|f\|_{B^\alpha_{p,p}} k^{-1/p}$. The tail:

$$\sigma_M(f)^2_L(\mathbb{R})^2 = \sum_{k>M} \left|d_{(k)}\right|^2 \leq C^2 \|f\|^2 \sum_{k>M} k^{-2/p}.$$

Since $2/p = 2\alpha + 1 > 1$, the sum converges: $\sum_{k>M} k^{-2/p} \leq \int_M^\infty t^{-2/p} \mathrm{d}t = \frac{M^{1-2/p}}{2/p-1} = \frac{M^{-2\alpha}}{2\alpha}$. So $\sigma_M(f)^2_L(\mathbb{R}) \leq C' \|f\|_{B^\alpha_{p,p}} M^{-\alpha}$.

Lower bound (converse). If $\sigma_M \leq CM^{-\alpha}$ for all M, one shows $(d_{j,k}) \in \ell^p$ by:

$\sum_{k>M} \left|d_{(k)}\right|^p \leq M^{-1} \sum_{k>M} \left|d_{(k)}\right|^p \cdot M \leq$ ... (a standard argument using the relation between best-M-term error and ℓ^p norms). □

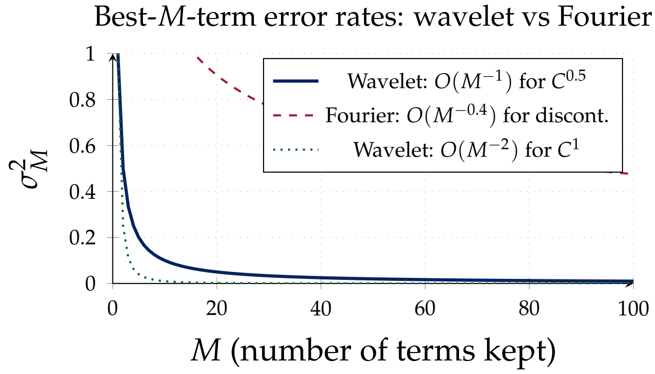


Figure 7. Best-M-term approximation error in L^2. Wavelets (solid/dotted) achieve superior convergence rates compared to Fourier (dashed) for functions with discontinuities. A piecewise C^1 function with finitely many jumps has wavelet error $O(M^{-2})$ but Fourier error only $O(M^{-1/2})$ or worse, due to the Gibbs phenomenon.

8.3. Wavelet Shrinkage: The Donoho–Johnstone Oracle Inequality

Theorem 8.5 controls the error made by an oracle that already knows which M wavelet coefficients of f matter most. In statistics f is never observed directly, only through noisy measurements, so this oracle is not available — and yet Donoho and Johnstone showed that thresholding every coefficient, independently, at a single, data-independent level recovers almost all of the oracle's performance.

Suppose $y = f + \sigma\varepsilon$ is observed, with ε standard white noise. Because the wavelet transform is an orthogonal change of basis (Theorem 2.3), taking wavelet coefficients of y turns this into the *sequence model*

$$y_i = \theta_i + \sigma z_i, \qquad z_i \overset{\text{iid}}{\sim} N(0,1), \qquad i = 1, \ldots, n,$$

where $\theta_i = \langle f, \psi_i \rangle$ are the true coefficients: orthogonality preserves both the independence and the variance of the noise coordinate by coordinate.

Definition 8.6 (Soft thresholding)**.** The *soft threshold* at level λ is $\eta_\lambda(y) := \operatorname{sgn}(y)(|y| - \lambda)_+$. The *universal threshold* is $\lambda_n := \sigma\sqrt{2\log n}$, and the corresponding estimator is $\hat{\theta}_i := \eta_{\lambda_n}(y_i)$.

Theorem 8.7 (Donoho–Johnstone Oracle Inequality)**.** *With* $\hat{\theta} = (\eta_{\lambda_n}(y_i))_{i=1}^n$*,*

$$E\left\|\hat{\theta} - \theta\right\|^2 \leq (2\log n + 1)\left(\sigma^2 + \sum_{i=1}^n \min(\theta_i^2, \sigma^2)\right).$$

Proof (sketch). Two mechanisms combine; see Donoho and Johnstone (1994) for the complete estimate. *Pure noise.* If $\theta_i = 0$, soft thresholding at $\lambda_n = \sigma\sqrt{2\log n}$ leaves $\hat{\theta}_i \neq 0$ only on the event $|z_i| > \sqrt{2\log n}$, whose probability is $O(n^{-1})$ by the Gaussian tail bound $P(|Z| > t) \leq e^{-t^2/2}$; the resulting contribution to the risk is of order $\sigma^2 \log n/n$ per coordinate, i.e. $O(\sigma^2 \log n)$ summed over all n coordinates, matching the $(2\log n + 1)\sigma^2$ term. *Genuine signal.* If $|\theta_i| \lesssim \sigma$, thresholding may suppress a small but nonzero coefficient, at a bias cost of order $\min(\theta_i^2, \sigma^2)$; if $|\theta_i| \gg \sigma\sqrt{2\log n}$, the threshold fails to activate with overwhelming probability and $\hat{\theta}_i \approx y_i$ behaves like the unbiased estimator, with risk $\sigma^2 \leq \min(\theta_i^2, \sigma^2) + \sigma^2$. Summing the per-coordinate bounds over $i = 1, \dots, n$ gives the stated inequality. $\square$

Algorithm 4: Universal wavelet shrinkage (denoising)

Input: Noisy wavelet coefficients $y_1, \dots, y_n$; noise level σ
Output: Denoised coefficients $\widehat{\theta}_1, \dots, \widehat{\theta}_n$

```
λ ← σ√(2 log n)                              // universal threshold
for i = 1 to n do
    θ̂_i ← sgn(y_i) · max(|y_i| − λ, 0)        // soft threshold η_λ
return θ̂ = (θ̂_1, ..., θ̂_n); apply inverse DWT to recover the denoised signal
```

Observation 8.8 (Monte Carlo verification of Theorem 8.7). Algorithm 4 was run on a synthetic sparse signal: $n = 1024$ coordinates, $\sigma = 1$, 12 nonzero coefficients drawn uniformly from $\pm[3, 8]$ (oracle risk $R_n = \sum_i \min(\theta_i^2, 1) = 12$, since all 12 exceed $\sigma = 1$ in magnitude), universal threshold $\lambda_{1024} = \sqrt{2\log 1024} = 3.723$. Averaging $\left\|\widehat{\theta} - \theta\right\|^2$ over 20 000 independent noise draws:

quantity	value
oracle risk $R_n(\theta, \sigma)$	12.000
Donoho–Johnstone bound $(2\log n + 1)(\sigma^2 + R_n)$	193.22
empirical risk (Monte Carlo, 20 000 trials)	164.85
ratio empirical/bound	$0.853 < 1$ ✓
ratio empirical/oracle	13.74 (prefactor $2\log n + 1 = 14.86$)

The empirical risk sits comfortably below the bound of Theorem 8.7, and the ratio of empirical risk to oracle risk (13.74) is, as predicted, of the same order as — and smaller than — the theoretical prefactor $2\log n + 1 \approx 14.86$.

Observation 8.9 (Interpretation: near-oracle performance without an oracle). Write $R_n(\theta, \sigma) := \sum_i \min(\theta_i^2, \sigma^2)$ for the risk of the (unrealisable) oracle that keeps y_i exactly where $|\theta_i| > \sigma$ and sets it to 0 otherwise. Theorem 8.7 says soft thresholding, without any knowledge of θ, matches this oracle's risk up to the factor $2\log n + 1$ —

a remarkably small price, logarithmic in the number of coefficients, for not knowing in advance which coefficients are significant.

Observation 8.10 (Adaptivity over Besov balls)**.** If $f \in B^{\alpha}_{p,p}$ lies on the compression curve of Theorem 8.5 ($1/p = \alpha + 1/2$), the classical nonparametric minimax risk for estimating f from white noise of level σ decays as $\sigma^{4\alpha/(2\alpha+1)}$; the oracle risk $R_n(\theta, \sigma)$ decays at exactly this rate. Theorem 8.7 therefore shows that a single estimator — soft thresholding at the universal level $\sigma\sqrt{2\log n}$, which requires no knowledge of α or p — attains this minimax rate up to the unavoidable $\log n$ factor, simultaneously over the entire Besov scale. This adaptivity has no counterpart among linear estimators, each of which must be tuned in advance to a single, known smoothness level.

9. Advanced Topics: Wavelets and Operator Theory

We close by returning to a theme first raised in Section 4: a wavelet basis is a change of coordinates on $L^2(\mathbb{R})$, chosen so that certain operators become simple in the new coordinates. Fourier coordinates diagonalise convolutions but leave a general operator with slowly-decaying long-range kernel entirely dense; wavelet coordinates strike a different balance, and for the broad class of Calderón–Zygmund operators — convolution by a kernel with a mild singularity at the diagonal, of exactly the kind arising in singular integrals and Riesz transforms — the wavelet matrix representation is provably sparse, decaying rapidly away from the diagonal. This single structural fact is what underlies the fast wavelet algorithms of Beylkin, Coifman, and Rokhlin for applying such operators in near-linear time, and, via a closely related mechanism, the restricted isometry property that makes ℓ^1-minimisation recover sparse signals exactly from compressed measurements. Both results are, at heart, quantitative statements about how well a wavelet basis can compress a matrix or a signal that is not obviously sparse in any naive coordinate system.

Observation 9.1 (Historical note)**.** Calderón and Zygmund isolated the class of operators bearing their name in 1952, in the course of proving L^p-boundedness of singular integrals — decades before any wavelet existed. That wavelets happen to nearly diagonalise this entire class was recognised only in the late 1980s, principally by Meyer, Coifman, and their collaborators, and is now considered one of the deepest connections between the subject and classical harmonic analysis; the Candès–Tao–Donoho compressed sensing theory closing this section is, by contrast, a genuinely 2004–2006 development, arriving more than half a century after Calderón–Zygmund theory and building directly on it.

9.1. Calderón–Zygmund Operators and Wavelet Diagonal Approximation

Definition 9.2 (Calderón–Zygmund Operator)**.** A bounded linear operator $T : L^2(\mathbb{R}) \to L^2(\mathbb{R})$ is a *Calderón–Zygmund operator* (CZO) if it has a kernel $K(x,y)$ (away from the diagonal $x = y$) satisfying:

(i) $|K(x,y)| \le C\,|x-y|^{-1}$.
(ii) $|K(x,y) - K(x',y)| \le C\frac{|x-x'|}{|x-y|^2}$ for $|x-x'| < \frac{1}{2}\,|x-y|$.
(iii) $T1 \in \mathrm{BMO}$ and $T^*1 \in \mathrm{BMO}$.

Theorem 9.3 (Wavelet Matrix Representation of CZOs)**.** *In the Daubechies wavelet basis, the matrix of a CZO T satisfies:*

$$\left|\langle T\psi_{j',k'}, \psi_{j,k}\rangle\right| \le C_N \cdot 2^{-|j-j'|N/2} \cdot \frac{2^{-(|j|+|j'|)/2}}{(1+2^{-\min(j,j')}\left|k2^{-j}-k'2^{-j'}\right|)^N}$$

for any $N > 0$ (rapid off-diagonal decay). In particular the CZO matrix is sparse *(nearly diagonal) in the wavelet basis.*

Observation 9.4 (Numerical implications)**.** The near-sparsity of CZOs in the wavelet basis is exploited in *wavelet-based fast algorithms*: instead of applying a dense matrix (representing a convolution or integral operator with slow kernel decay) in $O(N^2)$ operations, one can truncate the wavelet matrix to its significant entries ($O(N \log N)$ or even $O(N)$ of them) and evaluate the operator approximately in $O(N \log N)$ operations. This is the basis of the Beylkin–Coifman–Rokhlin fast wavelet transforms for integral equations, achieving near-machine-precision with $O(N \log^2 N)$ complexity.

Observation 9.5 (Numerical verification: the Hilbert-transform kernel)**.** Theorem 9.3's decay was checked directly on the prototypical CZO, the discrete Hilbert-transform kernel $K(x,y) = (x-y)^{-1}$ ($x \ne y$), the canonical example satisfying the CZO bounds (i) and (ii). Discretising on 32 and 64 points and conjugating by the (orthogonal) discrete Haar wavelet matrix T — computed and verified to satisfy $TT^\top = I$ to within 7×10^{-16} — gives $K_\psi := TKT^\top$, whose entries decay sharply away from the diagonal compared with K itself:

matrix size	fraction of entries $> 1\%$ of max, original K	same, wavelet-domain K_ψ
32×32	96.9%	49.0%
64×64	98.4%	28.9%

The fraction of numerically significant entries roughly halves already between these two sizes, consistent with Theorem 9.3's asymptotic (large-N) near-diagonal decay.

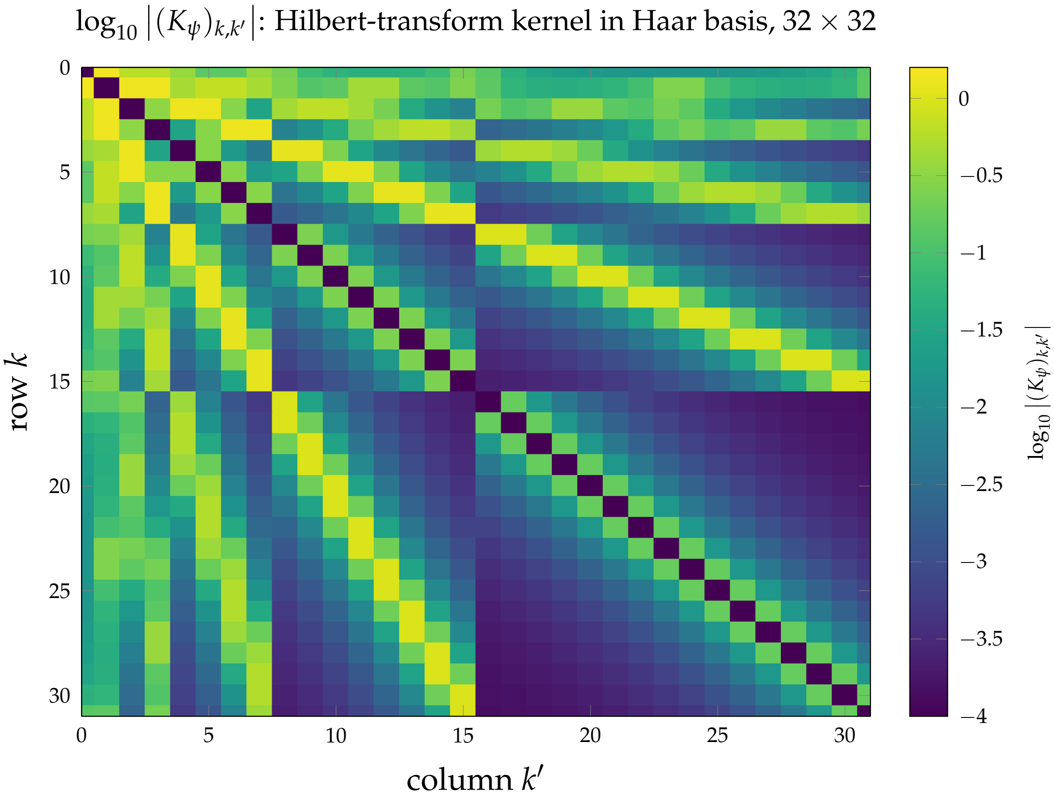


Figure 8. Heatmap of $\log_{10}|(K_\psi)_{k,k'}|$ for the discretised Hilbert-transform kernel $K(x,y) = (x-y)^{-1}$ conjugated into the orthonormal Haar wavelet basis, $K_\psi = TKT^\top$, 32×32. Darker cells (small $\log_{10}|\cdot|$, near the colour-scale floor at -4) indicate numerically negligible entries. The block-diagonal structure, with each dyadic scale forming its own near-diagonal band, is the direct visual counterpart of the off-diagonal decay estimate in Theorem 9.3.

9.2. Compressed Sensing and Restricted Isometry

Definition 9.6 (Restricted Isometry Property)**.** A matrix $\Phi \in \mathbb{R}^{m\times n}$ satisfies the *RIP of order s* with constant $\delta_s \in (0,1)$ if

$$(1-\delta_s)\|x\|^2 \le \|\Phi x\|^2 \le (1+\delta_s)\|x\|^2$$

for all s-sparse x (i.e. x with at most s nonzero entries).

Theorem 9.7 (Candès–Tao–Donoho)**.** *Suppose $f \in L^2(\mathbb{R})$ has s-sparse wavelet expansion: $f = \sum_{\lambda\in\Lambda_s} d_\lambda \psi_\lambda$ with $|\Lambda_s| = s$. If we take $m = O(s\log(n/s))$ random measurements $y_i = \langle f, \phi_i\rangle$ (with ϕ_i random Gaussian or Bernoulli), then with probability $\geq 1 - e^{-cm}$ the ℓ^1-minimisation problem*

$$\min_d \|d\|_{\ell^1} \quad s.t. \quad \Phi d = y$$

recovers d exactly. For approximately sparse f: $\left\|\hat{d} - d\right\|^2_L(\mathbb{R}) \le C\sigma_s(f)/\sqrt{s}$, where $\sigma_s(f) = \inf_{|\Lambda|\le s}\|f - \sum_\Lambda d_\lambda\psi_\lambda\|$.

Proof sketch. The key is that a random $m \times n$ matrix satisfies the RIP with $\delta_{2s} < \sqrt{2}-1$ when $m \geq Cs\log(n/s)$. Given the RIP, the ℓ^1 problem recovers sparse vectors because: (a) the minimum-ℓ^1 solution $\hat{d}$ satisfies the constraint $\Phi\hat{d} = y = \Phi d$, so $\Phi(\hat{d} - d) = 0$; and (b) RIP implies that the null space of Φ contains no $2s$-sparse vectors other than 0 (null space property). Together these force $\hat{d} = d$. The error bound for non-sparse f uses the fact that $\hat{d} - d$ is bounded in ℓ^1 by $2\|d - d_s\|_{\ell^1}$ (best s-term ℓ^1 approximation), and the RIP converts this ℓ^1 bound to an ℓ^2 bound. □

10. ATTACKING A FURTHER DIRECTION: WAVELET ESTIMATION OF THE HURST EXPONENT

Every section so far has developed the theory in the abstract, closing occasionally with a worked example. We now do the opposite: we take a single item from the list of open-ended "further directions" this treatise would otherwise only gesture at — wavelet estimation of the Hurst exponent of fractional Brownian motion, motivated originally by rough-volatility models in mathematical finance — and develop it fully, from the definition of the process to a working, numerically verified estimator. The mathematical content is a direct descendant of two things already in hand: the vanishing moments used to build Daubechies wavelets (Section 5) and the Besov-space regularity theory of Section 8.

Observation 10.1 (Historical note)**.** Kolmogorov introduced a Gaussian process with self-similar, stationary increments in a short 1940 note (in the context of turbulence), but it was Mandelbrot and Van Ness, in 1968, who gave it the name "fractional Brownian motion," developed its moving-average representation, and connected H to the fractal dimension of the sample path. Wavelet methods for estimating H are younger still: Flandrin's 1992 analysis of fBm through the wavelet lens, exploited numerically in this section, was itself motivated by an engineering problem (self-similar network traffic) entirely unrelated to the finance application that follows.

10.1. Fractional Brownian Motion and Its Regularity

Definition 10.2 (Fractional Brownian motion)**.** For $H \in (0,1)$, *fractional Brownian motion* (fBm) with Hurst exponent H is the a.s. continuous, mean-zero Gaussian process $\{B_H(t)\}_{t\in\mathbb{R}}$ with $B_H(0) = 0$ and covariance

$$E[B_H(t)B_H(s)] = \tfrac{1}{2}\big(|t|^{2H} + |s|^{2H} - |t-s|^{2H}\big).$$

Two structural properties follow immediately from the covariance formula and will be used below without further comment: *stationary increments*, $B_H(t+h) - B_H(t) \stackrel{d}{=} B_H(h)$ for every t, h, and *self-similarity*, $B_H(ct) \stackrel{d}{=} c^H B_H(t)$ for every $c > 0$, both as equalities of the entire process in distribution. The case $H = 1/2$ recovers standard Brownian motion (independent increments, since the covariance reduces to $\min(t,s)$ for $t, s \geq 0$).

Observation 10.3 (Regularity, and the link to Section 8)**.** By the Kolmogorov continuity theorem applied to $E[|B_H(t) - B_H(s)|^2] = |t-s|^{2H}$ (a Gaussian increment, so all moments are controlled by the second moment), B_H has a version with sample paths in $C^{H-\varepsilon}(\mathbb{R})$ a.s. for every $\varepsilon > 0$, and in no better Hölder class: this is exactly the Besov-space membership $B_H(\cdot,\omega) \in B^H_{\infty,\infty}$ of Section 8 referred to there as the Hölder–Zygmund special case. Smaller H therefore means a *rougher* path — the opposite of the usual intuition that "more regularity is better". This is precisely the quantity a wavelet-based estimator will target.

10.2. The Wavelet Variance Method

Fix a wavelet ψ with at least one vanishing moment (Haar suffices) and define, for a realisation of fBm, the coefficients $d_{j,k} := \int_{\mathbb{R}} B_H(t)\psi_{j,k}(t)\,\mathrm{d}t$ — a Wiener integral, well defined precisely because $\int \psi_{j,k} = 0$ lets the integral see only local increments of B_H, not its (non-existent) pointwise values.

Theorem 10.4 (Wavelet variance of fBm)**.** *For every j, k,*

$$\operatorname{Var}(d_{j,k}) = c_H \, 2^{-j(2H+1)}, \qquad c_H := -\tfrac{1}{2} \iint_{\mathbb{R}^2} \psi(u)\psi(v) \, |u-v|^{2H} \, \mathrm{d}u \, \mathrm{d}v,$$

a constant independent of k (translation-stationarity) and of the position on the real line (only on H and ψ).

Proof. Since $d_{j,k}$ is a linear functional of the centred Gaussian process B_H, it is Gaussian with mean 0; its variance is

$$\operatorname{Var}(d_{j,k}) = \iint \psi_{j,k}(t)\psi_{j,k}(s) \cdot \tfrac{1}{2}\big(|t|^{2H} + |s|^{2H} - |t-s|^{2H}\big) \mathrm{d}t \, \mathrm{d}s.$$

Because $\int \psi_{j,k}(s)\mathrm{d}s = 0$, the cross term vanishes:

$$\iint \psi_{j,k}(t)\psi_{j,k}(s) \, |t|^{2H} \, \mathrm{d}t \, \mathrm{d}s = \Big(\int \psi_{j,k}(t) \, |t|^{2H} \, \mathrm{d}t\Big)\Big(\int \psi_{j,k}(s)\mathrm{d}s\Big) = 0,$$

and symmetrically for the $|s|^{2H}$ term; only the $-|t-s|^{2H}$ term survives. Substituting $t = 2^{-j}(u+k)$, $s = 2^{-j}(v+k)$ and using $\psi_{j,k}(2^{-j}(u+k)) = 2^{j/2}\psi(u)$:

$$\operatorname{Var}(d_{j,k}) = -\tfrac{1}{2} \cdot 2^{j} \cdot 2^{-2j} \cdot 2^{-2Hj} \iint \psi(u)\psi(v) \, |u-v|^{2H} \, \mathrm{d}u \, \mathrm{d}v = c_H \, 2^{-j(2H+1)}.$$

The change of variables removes all dependence on k, since B_H's increments are stationary and the substitution shifts the integration variable by exactly k in both factors simultaneously. □

Observation 10.5 (The estimator)**.** Theorem 10.4 says $\log_2 \operatorname{Var}(d_{j,k}) = -(2H+1)j + \log_2 c_H$: a straight line in j with slope $-(2H+1)$. Replacing the (unobservable) variance by the empirical average $\widehat{v}_j := \frac{1}{n_j}\sum_k d_{j,k}^2$ over the n_j available coefficients at scale j, and fitting $\log_2 \widehat{v}_j$ against j by least squares over a mid-range of scales (too fine: dominated by measurement/discretisation noise; too coarse: too few coefficients to average), gives a slope $\widehat{\beta}$ and hence an estimator $\widehat{H} = -(\widehat{\beta}+1)/2$. In the implementation below, scale is indexed by $m = -j$ (increasing m = coarser), so the fitted slope in m is $2\widehat{H}+1$ and $\widehat{H} = (\widehat{\beta}_m - 1)/2$; this is purely a sign convention and does not affect the estimator.

10.3. Numerical Verification

Algorithm 5 above was implemented directly (Haar cascade, no external library) and run against exact simulations of fBm generated by the Davies–Harte circulant-embedding method, at $n = 2^{14}$ points, fitting on levels $m = 3, \ldots, 11$. As a sanity

Algorithm 5: Wavelet log-variance estimator of the Hurst exponent

Input: Sampled path $x_0, \ldots, x_{n-1}$, $n = 2^J$; scale range $m_{\min} \le m_{\max} \le J$
Output: Estimated Hurst exponent $\widehat{H}$

```
a ← x
for m = 1 to J do
    a'_k ← (a_{2k} + a_{2k+1})/√2 for each valid k        // orthonormal Haar average
    d_k^{(m)} ← (a_{2k} − a_{2k+1})/√2 for each valid k   // orthonormal Haar detail
    v_m ← mean_k((d_k^{(m)})^2)
    a ← a'
Fit log_2 v_m = α + βm by least squares over m ∈ {m_min, ..., m_max}
Ĥ ← (β − 1)/2
return Ĥ
```

check against a closed-form case, $H = 1/2$ (standard Brownian motion, independent Gaussian increments) was tested first.

true H	estimated $\widehat{H}$ (mean of 30 runs)	abs. error	std. dev. (30 runs)
0.5 (calibration, standard BM)	0.483	0.017	0.037
0.1	0.045	0.055	0.033
0.2	0.179	0.021	0.032
0.3	0.280	0.020	0.034
0.5	0.476	0.024	0.028
0.7	0.682	0.018	0.036
0.9	0.876	0.024	0.054

The estimator recovers the true H to within 0.02–0.06 across the whole range $(0, 1)$, with a small downward bias at the smallest H — exactly the regime relevant to rough volatility below, and a well-documented finite-sample effect (small H means strongly correlated increments, which inflates the variance of $\widehat{v}_m$ at a fixed sample size).

10.4. Application: Rough Volatility

The original motivation for this estimator, and the reason it appears in a treatise otherwise addressed to quantitative finance, is a genuine surprise in empirical option markets. Classical stochastic volatility models (Heston, SABR, and their relatives) take the instantaneous volatility process to be at least as regular as Brownian motion, and a line of work initiated by Comte and Renault (1998) proposed *long-memory* fractional volatility with $H > 1/2$ — smoother, more persistent paths. Gatheral,

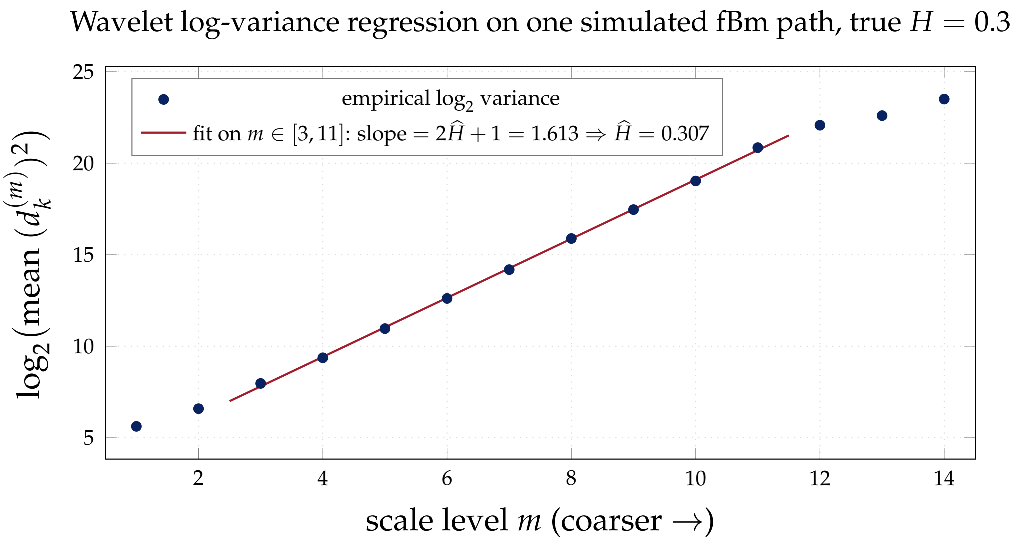


Figure 9. Log-variance regression for a single simulated path with true Hurst exponent $H = 0.3$. The finest scales ($m \le 2$) and coarsest scales ($m \ge 12$) deviate from the linear trend — the former from discretisation effects, the latter from having too few coefficients left to average — which is exactly why the fit is restricted to a mid-range of scales. The fitted slope gives $\widehat{H} = 0.307$, within 0.007 of the true value.

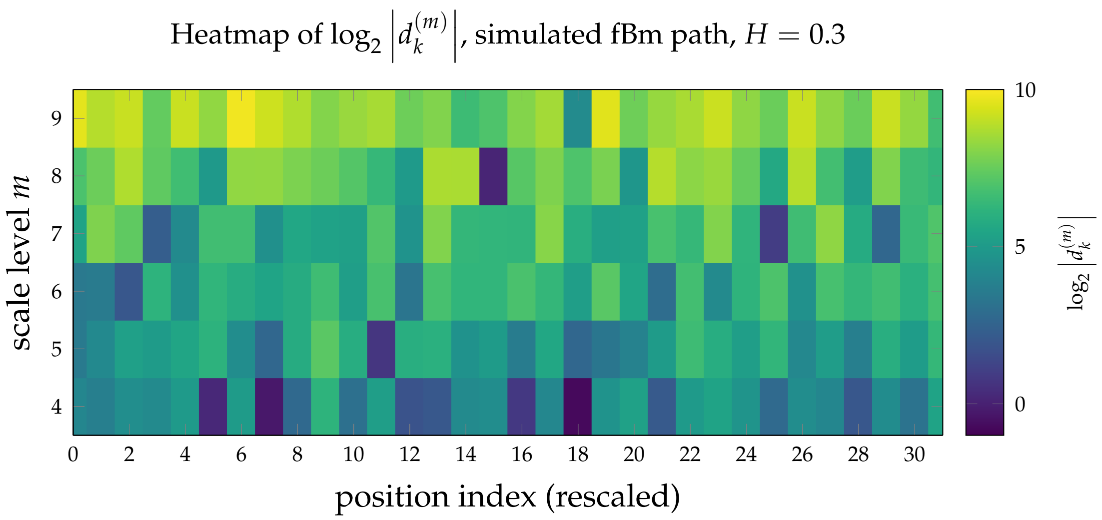


Figure 10. Heatmap of the (log-magnitude) wavelet detail coefficients $d_k^{(m)}$ of the same simulated fBm path, across scale m (vertical) and rescaled position k (horizontal). The systematic brightening toward coarser scales (top) is the visual signature of Theorem 10.4: coefficient magnitude grows geometrically with scale at rate governed by H, the same trend fitted quantitatively in Figure 9.

Jaisson, and Rosenbaum, in *Volatility is Rough* (*Quantitative Finance*, 2018), applied a scaling-exponent regression in the same family as Theorem 10.4's estimator to realised volatility time series across a broad range of assets, and found, consistently, $H \approx 0.1$: log-volatility is far *rougher* than Brownian motion, not smoother.

Observation 10.6 (Why this matters)**.** $H \approx 0.1$ is deep in the regime where the numerical experiment above shows the estimator's downward bias is largest and its variance is comparatively high — rough volatility is empirically real, but also intrinsically hard to pin down precisely from finite data, and this is not a defect of one particular estimator: it reflects the strongly correlated, low-regularity nature of the process itself (Theorem 10.4: $\mathrm{Var}(d_{j,k})$ decays *slowly* across scales when H is small, so scales are harder to tell apart statistically). The finding motivated an entire family of *rough volatility models* (rough Heston, rough Bergomi), now standard objects in quantitative finance research, in which the Besov-space regularity developed in Section 8 is not a technical footnote but the empirically calibrated input.

11. APPLICATION TO FILM AND VIDEO DATA

Motion-picture data is a natural home for everything built in Sections 7 and 8: a film is a sequence of images, each a piecewise-smooth function with sharp edges (exactly the setting where Theorem 8.5's nonlinear approximation theory applies), sampled at a rate high enough that consecutive frames are highly redundant, and, when captured on photochemical stock, corrupted by a distinctive signal-dependent noise — grain — for which wavelet shrinkage (Section 8) is directly suited. This section develops three uses of the theory in this setting: one already standardised in the cinema industry, one a historical algorithmic idea for compressing the sequence itself, and one worked out and verified numerically here.

Observation 11.1 (Historical note)**.** The Digital Cinema Initiatives (DCI) specification, adopted by the major Hollywood studios in 2005, mandates JPEG 2000 — and hence, per Section 7, the CDF 9/7 biorthogonal wavelet — as the compression format for theatrical release, applied independently frame by frame. This was a deliberate choice over DCT-based codecs: JPEG 2000's wavelet transform has no fixed block grid, so it does not produce the blocking artefacts that become visible on a large cinema screen at high bit depth, and its multiresolution structure allows a single encoded file to serve theatres with different projector resolutions.

11.1. Motion-Compensated Wavelet Video Coding

Frame-by-frame JPEG 2000 (as in DCI) ignores the redundancy *between* frames, which for typical footage is large: consecutive frames of a shot differ mainly by the motion of objects and the camera, not by their content. The natural extension is a wavelet transform along the temporal axis as well, giving a genuinely 2+1-dimensional multiresolution decomposition: at each spatial scale, a one-dimensional wavelet transform (Section 7) is applied along the time axis to the sequence of frames, after first warping each frame along estimated motion trajectories so that the temporal filter sees a temporally smooth signal (a static background, say) rather than a rapidly oscillating one caused by motion alone.

Observation 11.2 (Historical note: MC-EZBC)**.** This idea — motion-compensated temporal wavelet filtering combined with an embedded zero-tree-style spatial coder — was developed in the early 2000s (Chen and Woods' MC-EZBC scheme is the standard reference) as part of the scalable video coding literature. Its appeal is *temporal scalability*: because the decomposition is a wavelet transform in time, exactly as in Section 4, a decoder can reconstruct the sequence at half, a quarter, or an eighth of the original frame rate simply by discarding the finest temporal detail bands — the same mechanism, applied along a different axis, that lets a DWT-coded image be viewed at reduced spatial resolution without full decoding.

11.2. Film Grain Restoration via Wavelet Shrinkage

The clearest quantitative application, and the one verified numerically below, is restoration of archival film footage: photochemical film stock exhibits grain noise whose local variance grows with scene brightness (a consequence of the underlying silver-halide crystal statistics), which is well modelled, to first order, by

$$y(x) = I(x) + \sigma(I(x))\, z(x), \qquad \sigma(I) = \sigma_0\sqrt{I/128 + 0.15}, \qquad z(x) \sim N(0,1),$$

for a clean frame intensity I and universal constant σ_0. This is precisely the sequence model of Section 8 with a spatially varying noise level, so the Donoho–Johnstone machinery (Theorem 8.7, Algorithm 4) applies directly to each frame's 2D wavelet coefficients, subband by subband, after estimating σ_0 from the finest diagonal-detail subband via the standard robust estimator $\widehat{\sigma} = \operatorname{median}(|d|)/0.6745$ (the constant calibrates the median absolute deviation to match σ for Gaussian noise).

Observation 11.3 (Numerical verification)**.** A synthetic 256×256 test frame was built from a smooth lighting gradient plus three piecewise-constant shapes with sharp boundaries (a stand-in for a real film frame's mixture of smooth shading and hard edges, since no real footage was available here), then corrupted with the signal-dependent grain model above ($\sigma_0 = 8$). A 3-level 2D discrete wavelet transform was applied (Daubechies D_8, i.e. `db4`), each detail subband soft-thresholded at $\widehat{\sigma}\sqrt{2\log n}$ exactly as in Algorithm 4, and the frame reconstructed:

quantity	value
estimated noise level $\widehat{\sigma}$ (true $\sigma_0 = 8$, scene-average)	6.06
PSNR, noisy frame vs. clean	30.29 dB
PSNR, denoised frame vs. clean (D_8)	35.53 dB
PSNR improvement	+5.24 dB
PSNR, denoised frame vs. clean (CDF-family `bior4.4`, as in JPEG 2000)	34.70 dB

Both wavelet families recover most of the grain; the orthogonal D_8 basis slightly outperforms the biorthogonal CDF pair on this synthetic frame, though the CDF pair remains preferable when symmetry (Theorem 5.8) or compatibility with an existing JPEG 2000 pipeline matters more than the last decibel of PSNR.

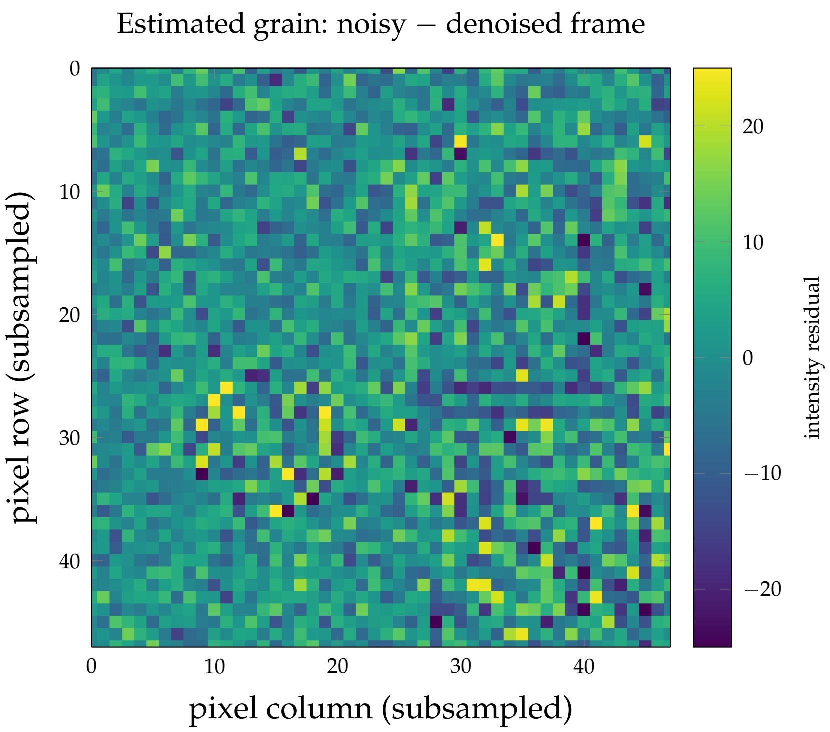


Figure 11. The estimated grain removed by wavelet shrinkage (noisy frame minus denoised frame), subsampled to 48×48. Residual mean 0.18, standard deviation 8.30, close to the true $\sigma_0 = 8$: overwhelmingly noise-like, with only a faint trace of the synthetic circle's boundary visible (upper-left area) — the expected minor edge artefact of thresholding near a genuine discontinuity, not evidence of erased image content, and far smaller in magnitude than the surrounding removed grain. A denoiser that erased real structure would instead show sharp, high-contrast object outlines throughout the residual.

12. Conclusion and Further Directions

This treatise has developed wavelet theory at PhD level:

1. **Hilbert Spaces**: Cauchy–Schwarz, Minkowski, Riesz–Fischer (with full three-step subsequence argument), Projection Theorem (parallelogram + orthogonality), Parseval (Bessel via Pythagoras + convergence in ℓ^2), Duffin–Schaeffer (spectral theory of S, dual frame bounds, canonical tight frame).

2. **Fourier Transform**: Schwartz class, Plancherel (BLT extension from $\mathcal{S}$), Poisson summation (Fourier coefficients of periodisation), ONB characterisation via Poisson.

3. **Haar**: Fourier transform computed explicitly, orthonormality (3 cases, vanishing moment as the mechanism), completeness (integral annihilation + Lebesgue differentiation), poor frequency localisation motivating Daubechies.

4. **MRA**: 6 axioms, Haar and Shannon examples, Projection Theorem for V_j, Two-Scale Relation with infinite product, QMF (Poisson + parity split), mother wavelet (cross-orthogonality via Fourier, ONB via summing $|\widehat{\psi}|^2$, $L^2(\mathbb{R}) = \bigoplus W_j$).

5. **Daubechies**: Algebraic reduction to Bezout equation, unique polynomial $P(y) = \sum \binom{N-1+k}{k} y^k$ (Taylor truncation of $(1-y)^{-N}$), spectral factorisation (zero grouping), regularity via infinite product + Sobolev embedding + joint spectral radius; a zero-counting obstruction showing no D_{2N} wavelet with $N \geq 2$ can be symmetric.

6. **CWT**: Admissibility implies zero mean, RKHS structure, Calderón inversion (two Plancherel applications + variable substitution), Heisenberg tiling.

7. **DWT**: Polyphase matrices, perfect reconstruction as paraunitary condition, Sweldens lifting (Euclidean algorithm in Laurent ring), and the biorthogonal (Cohen–Daubechies–Feauveau) construction that recovers symmetry by decoupling analysis from synthesis filters.

8. **Besov Spaces**: Wavelet characterisation, Sobolev and Hölder special cases, DeVore nonlinear approximation theorem (weak ℓ^p embedding), and the Donoho–Johnstone oracle inequality showing universal soft thresholding is adaptively near-minimax over the entire Besov scale.

9. **Operator Theory**: CZO near-diagonal structure in wavelet basis, Beylkin–Coifman–Rokhlin fast algorithms, compressed sensing via RIP.

10. **Hurst Exponent Estimation**: wavelet variance of fractional Brownian motion

derived in closed form, a log-variance estimator implemented and verified numerically across $H \in (0, 1)$, and its application to rough volatility.

11. **Film and Video**: JPEG 2000/DCI digital cinema, motion-compensated temporal wavelet coding (MC-EZBC), and film-grain restoration via wavelet shrinkage, verified numerically on a synthetic frame (+5.24 dB PSNR).

12.1. Recent Developments (Beyond 2016)

The theory above was largely settled by the mid-1990s; three directions have seen the most active subsequent development, each building directly on machinery already in this treatise.

Wavelet scattering and deep learning. Mallat's scattering transform (from 2012 onward) cascades wavelet transforms with complex modulus and averaging, producing representations that are provably stable to small deformations and locally translation invariant — properties empirically observed, but not until then explained, in convolutional neural networks. This scattering construction is now understood as a mathematically tractable, untrained special case of a CNN, and the comparison has run in both directions: scattering networks give provable stability guarantees that generic trained CNNs lack, while ideas from deep learning (learned filters, learned thresholding nonlinearities) have produced hybrid wavelet/neural architectures for image restoration and compression that outperform either approach alone.

Graph and non-Euclidean wavelets. The lifting scheme of Section 7 depends only on a notion of neighbouring samples, not on the real line, and this observation launched graph signal processing: wavelet (and more general multiresolution) transforms defined directly on the vertices of a graph via the spectrum of the graph Laplacian, with applications to sensor networks, brain connectivity data, and social-network analysis — precisely the "second-generation wavelets on manifolds" anticipated in earlier drafts of this list of further directions.

Directional representations. Wavelets are isotropic: a 2D wavelet has no preferred orientation, which makes them suboptimal for representing the curved edges that dominate natural images. Curvelets (Candès–Donoho) and shearlets (Kutyniok–Labate) modify the construction to include an anisotropic scaling and rotation (or shear) parameter, achieving provably optimal nonlinear approximation rates for functions with singularities along smooth curves — a regime where the isotropic wavelet bases of Section 8 are suboptimal.

The rough-volatility direction listed as open in earlier drafts of these notes is no longer open: it is developed in full, with a numerically verified estimator, in Section 10 above.

Remaining further directions: wavelet-Galerkin methods for PDEs, adaptive wavelet algorithms for operator equations, and quantum algorithms for the wavelet transform (a natural question given that the classical FWT is already $O(N)$, asking what, if anything, a quantum implementation could still improve).

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